DSAA 4011 · LECTURE 7 · THURSDAY 22 OCTOBER 2026

Who is linked to whom?

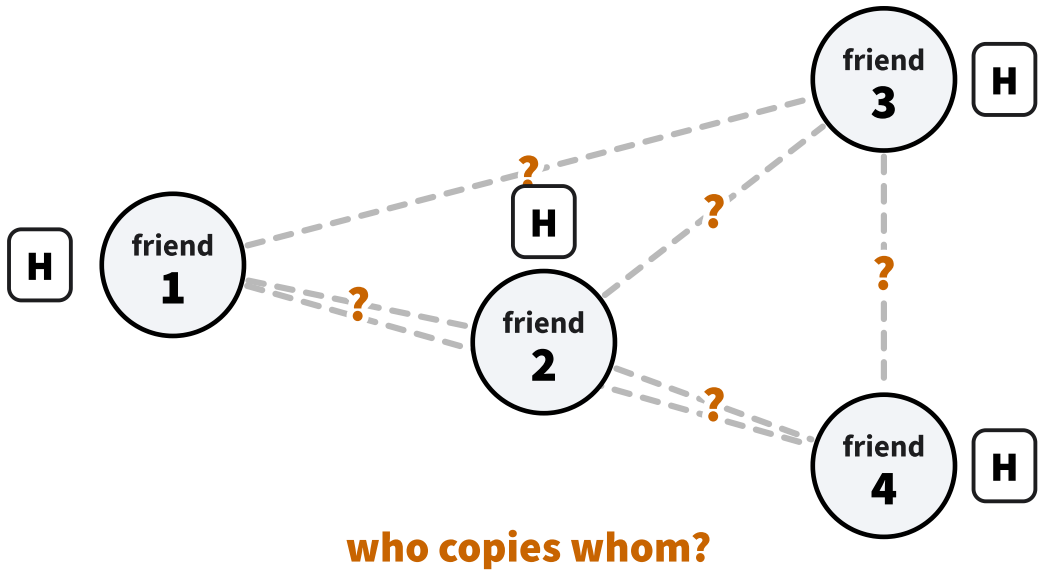
Learning the links from data: mutual information and Chow–Liu trees

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THE STORY

Four friends toss coins, and some copy each other

Each day, four friends toss coins and write the faces in a **notebook**. Some friends **copy** another friend's face, most of the time.



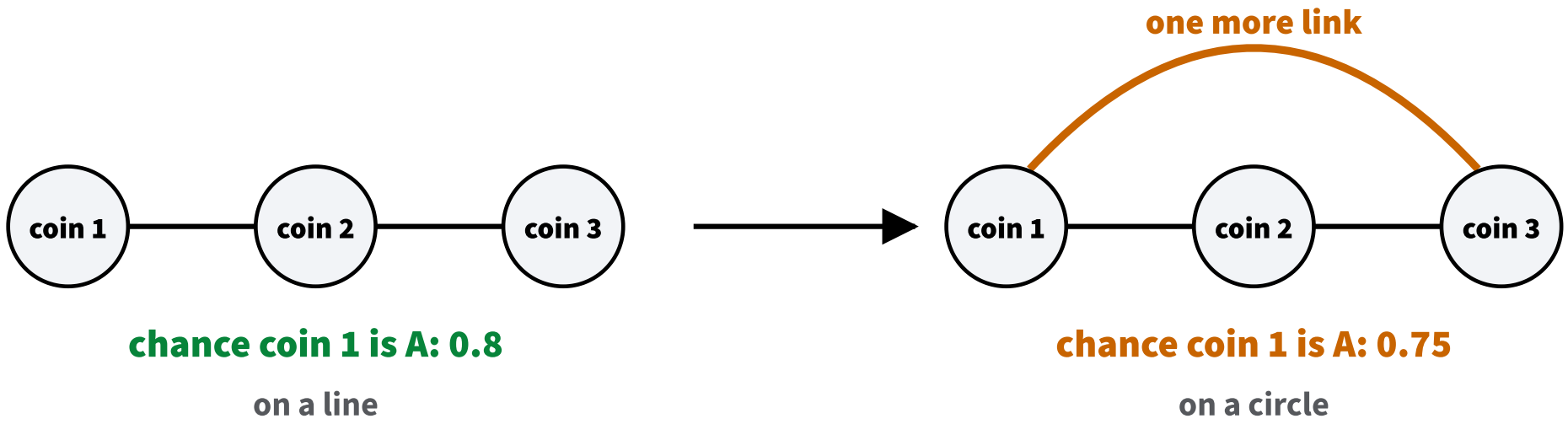
day	friend 1	friend 2	friend 3	friend 4
1	H	H	H	H
2	H	H	H	T
3	H	H	H	H
4	H	H	H	H
5	H	T	T	T
6	H	H	T	T
		...		

We see only the notebook. **Which friends copy each other?**

The links are hidden; we must learn them from the notebook.

Last week, the links were given, and they mattered

In Lecture 6 the links were **given**: a line or a circle. One more link changed every answer **(Lecture 6, pp. 60 and 65)**.

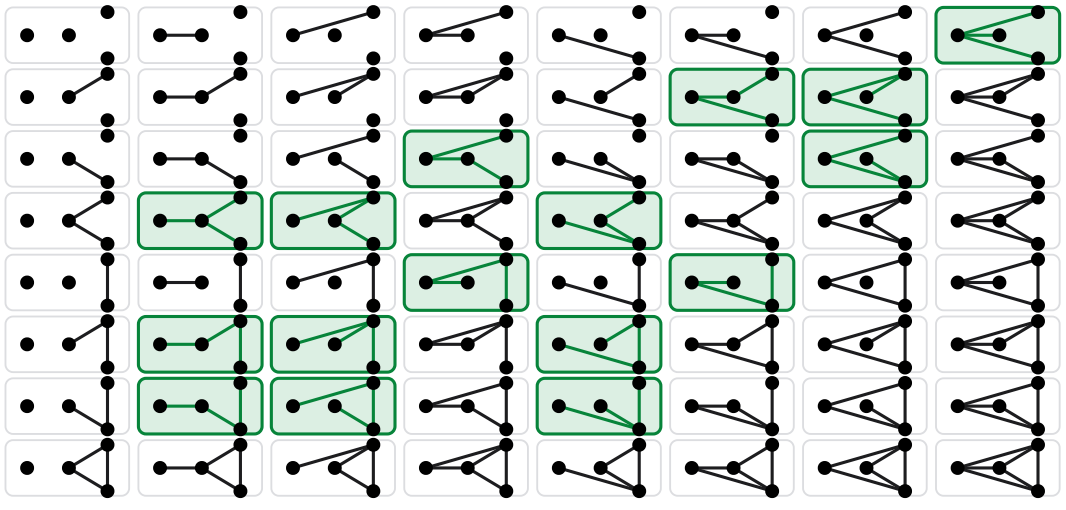


Today each circle is a **friend**, every face is seen, and we **learn the links from data**.

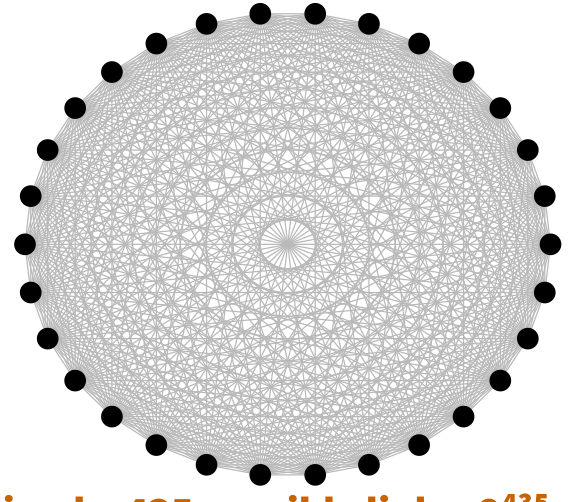
> **Every answer depends on the links, so the links must be right.**

There are far too many maps of links to try them all

A **map** says which pairs are linked. Each pair is in or out, so maps multiply.



4 friends: 6 possible links, 64 maps; 16 are trees (green)



30 friends: 435 possible links, 2^{435} maps

Today's question: how do we find the links **without trying every map**?

> **Four friends: try them all. Thirty friends: impossible.**

WARM-UP

Two ideas from earlier lectures that come back today

1. Two friends toss without looking at each other. What is the mutual information of their faces (Lecture 3)?

zero | ln 2 | one

Answer. **Zero:** one face tells nothing about the other. Today it **measures a link**.

2. In Lecture 5, with every coin seen, how did we fit a heads chance?

by expectation–maximization | heads ÷ flips | by sampling

Answer. **We counted:** heads ÷ flips. Today it **fits each link of a map**.

Information measures a link; counting fits it.

TODAY'S PLAN

Measure, score, build, check, then go beyond trees



from the notebook alone, to a map of links we can trust

Four steps take us from a notebook to a map we can trust.

OBJECTIVES

By the end of today, you can do six things

1. **measure** how strongly two friends are linked — **mutual information**
2. **score** a whole tree map from its links — **the score**
3. **build** the best tree, and prove it is best — **Chow–Liu**
4. **decide** whether to trust a learned link — **checks**
5. **test** a link given the other friends — **beyond trees**
6. **explain** which arrows the data can show — **colliders**

> **One notebook, four friends, one map to find.**

TODAY'S SYMBOLS

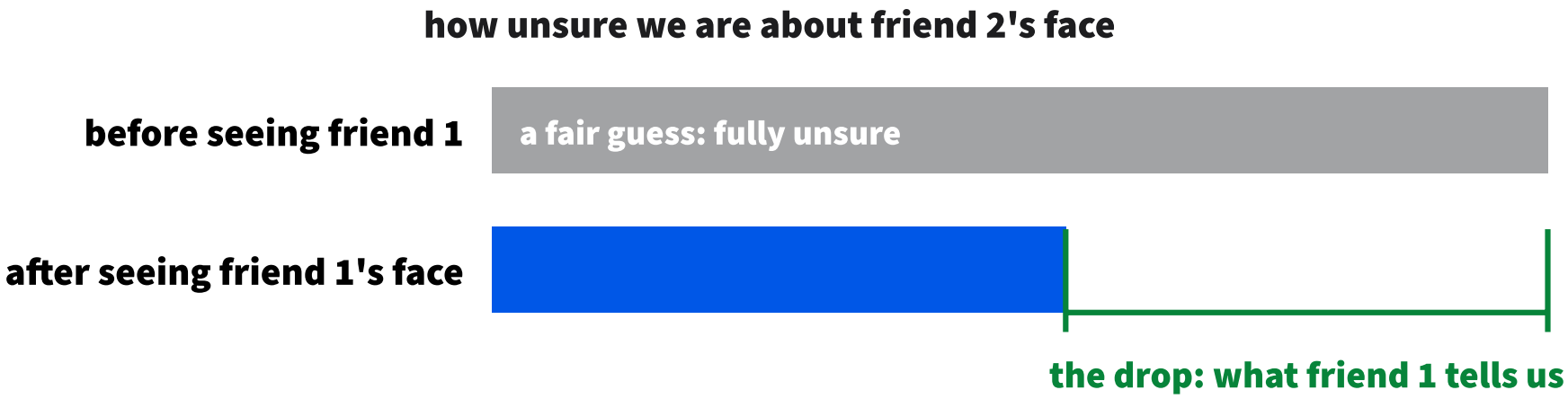
Each symbol today, with its meaning in words

symbol	meaning
x_i	the face of friend i on one day: H or T (upright letters are faces)
$\boldsymbol{x} = (x_1, x_2, x_3, x_4)$	one day: all four faces
$X_i, \boldsymbol{X}$	friend i's face and all four faces, as random quantities
$N, n(\cdot)$	days in the notebook; how many days show the faces in the brackets
$\hat{p}$	a counted fraction (hat = counted from the notebook), e.g. $\hat{p}(x_1{=}\mathrm{H}) = n(x_1{=}\mathrm{H})/N$
$H(\cdot), I(\cdot\,;\cdot)$	entropy and mutual information (Lecture 3), in nats
m	the number of friends: 4 today
$\mathcal{T}, p_{\mathcal{T}}$	a tree map (a set of links) and the model it defines

A hat means "counted from the notebook".

If friend 2 copies friend 1, one face predicts the other

Guess friend 2's face: **fully unsure**. See **friend 1's face** first: if 2 copies 1, the guess is **much easier**.

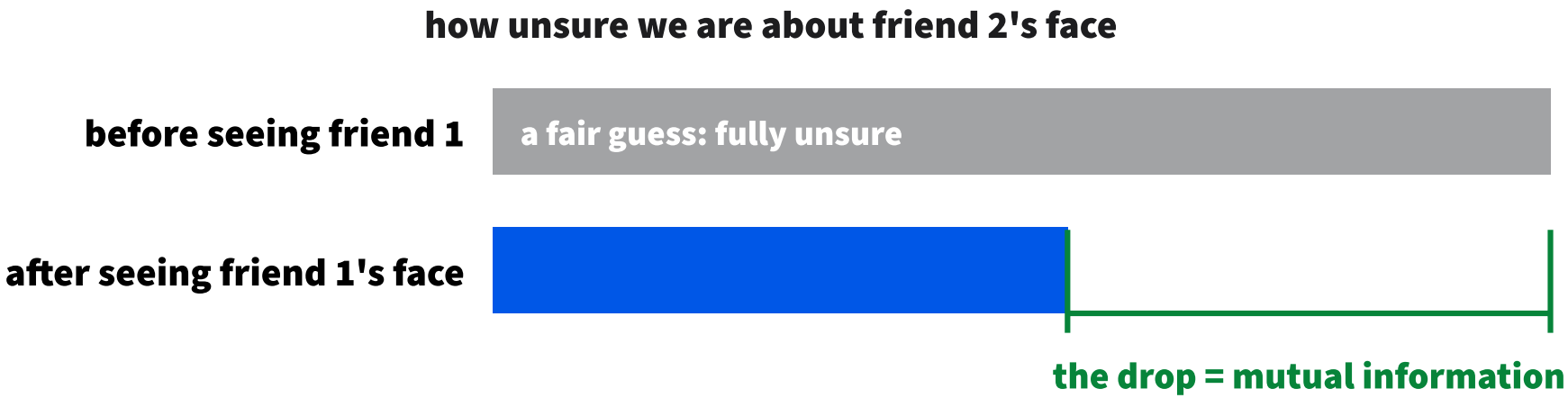


A link is strong when one face makes the other easy to guess.

The drop in uncertainty is the mutual information

1. **Before:** entropy $H(X_2)$ of friend 2's face.
2. **After:** $H(X_2 \mid X_1)$, once friend 1's face is seen.

Mutual information: $I(X_1; X_2) = H(X_2) - H(X_2 \mid X_1)$, in nats (Lecture 3, p. 59)



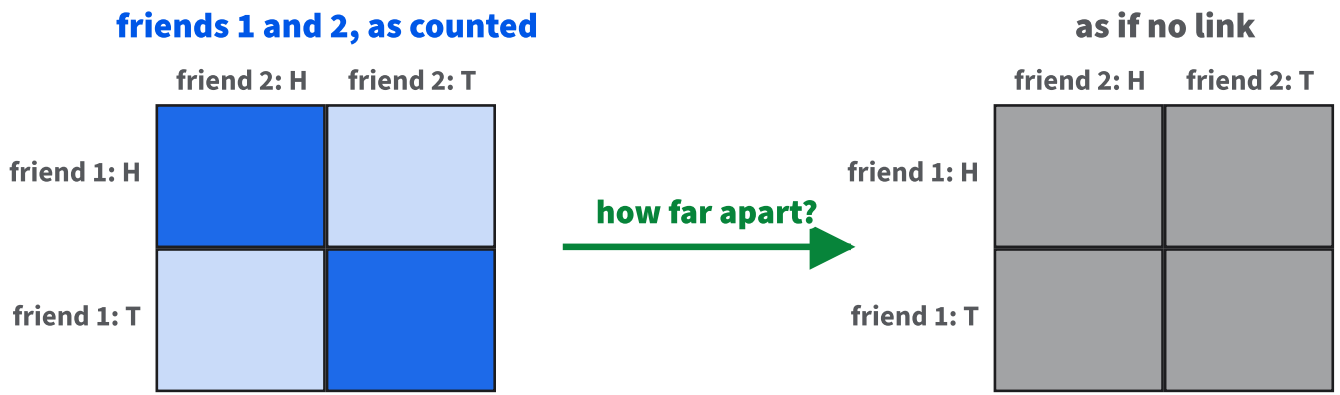
Mutual information: how much one face tells about the other.

Mutual information is the distance from no link

Compare the pair's table with the **no-link** table: the product of each face's chances.

$$I(X_1; X_2) = \sum_{x_1, x_2} p(x_1, x_2) \cdot \ln \frac{p(x_1, x_2)}{p(x_1) \cdot p(x_2)} = \mathrm{KL}\big(p(x_1, x_2) \,\|\, p(x_1) \cdot p(x_2)\big)$$ (Lecture 3, p. 60)

1. **Symmetric:** 1 tells about 2 what 2 tells about 1.
2. **Never negative**, like every Kullback–Leibler (KL) divergence.
3. **Zero exactly when there is no link:** the faces are independent.



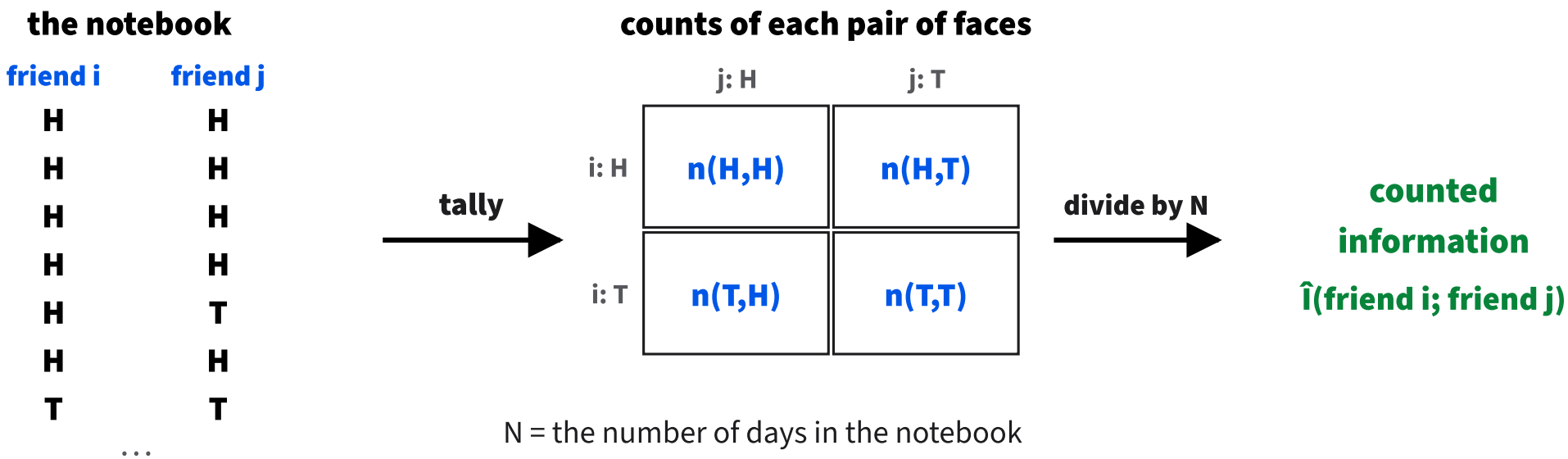
darker = more days; the counted table puts its weight where the faces agree

Zero means no link; larger means a stronger link.

From the notebook, we count each pair's table

True chances are unseen, so we **count**: $\hat{p}(x_i, x_j) = n(x_i, x_j)/N$, the **counted fraction**.

Counted information: $\hat{I}(X_i; X_j) = \sum_{x_i, x_j} \hat{p}(x_i, x_j) \cdot \ln \frac{\hat{p}(x_i, x_j)}{\hat{p}(x_i) \cdot \hat{p}(x_j)}$

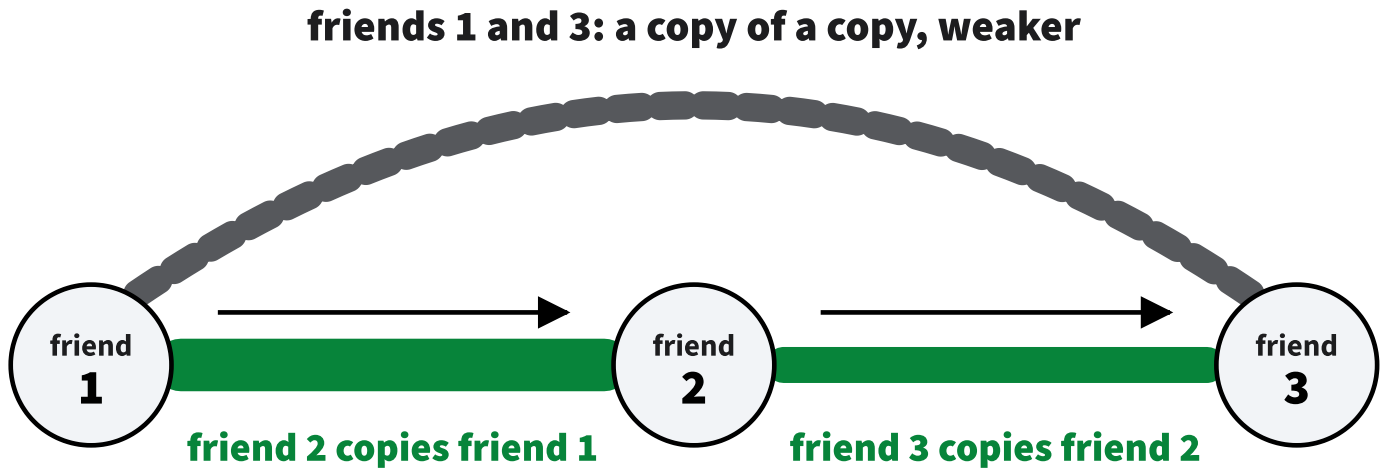


N = the number of days in the notebook

Counted fractions in, counted information out.

Copying a copy loses more: links fade along a chain

3 copies 2, who copied 1. **Each copy can go wrong**, so friend 3 tells **less** about friend 1 than friend 2 does.



line thickness = how strongly the faces are linked

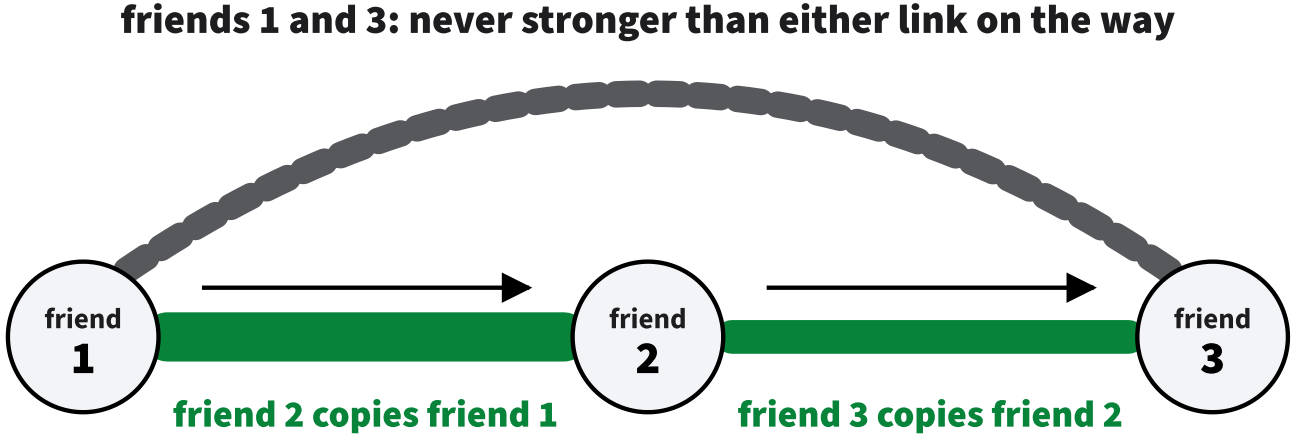
The indirect pair 1–3 is weaker than both links on its way.

Theorem A: information can only fade along a chain

Theorem A (data processing). If friend 3 depends on friend 1 only via friend 2 (Markov chain $X_1 \to X_2 \to X_3$), then $I(X_1; X_3) \le I(X_1; X_2)$ and $I(X_1; X_3) \le I(X_2; X_3)$.

In words: an indirect pair is never stronger than a link on its path.

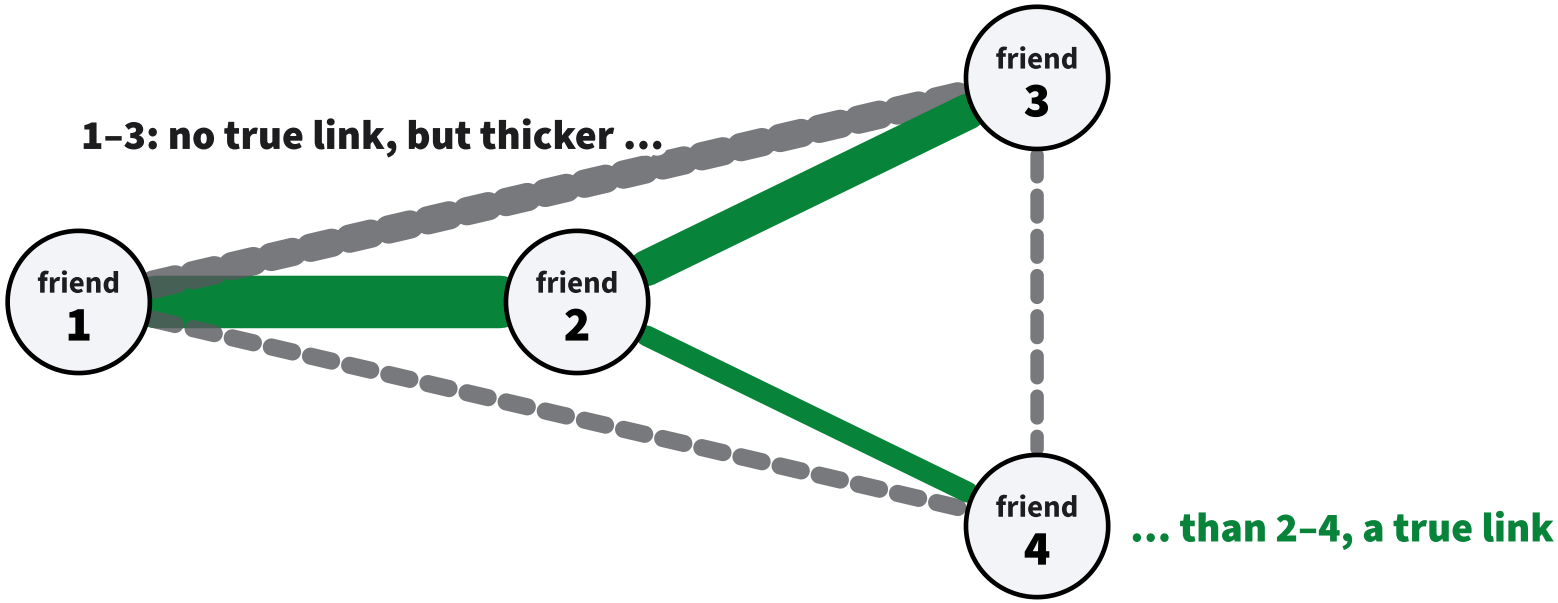
Proof: chain rule in both orders; friend 3 adds nothing once friend 2 is known, so $I(X_1; X_3) \le I(X_1; X_2)$ (Lecture 7, p. 95).



Remember: indirect pairs never beat a link on their path.

But an indirect pair can beat a weak true link

Theorem A bounds 1–3 only by **1–2 and 2–3**, its own path. So 1–3 can beat **2–4**, a **true but weak** link.



green: true links · grey dashed: pairs linked only through others · thickness: strength

Keeping the **three strongest pairs** links 1–3, misses 2–4.

> **The strongest pairs are not the true links.**

Next: score a whole map, not one pair at a time

Links must explain the notebook **together**: **one number per map**.



from the notebook alone, to a map of links we can trust

A tree map's score is a sum. A sum of what?

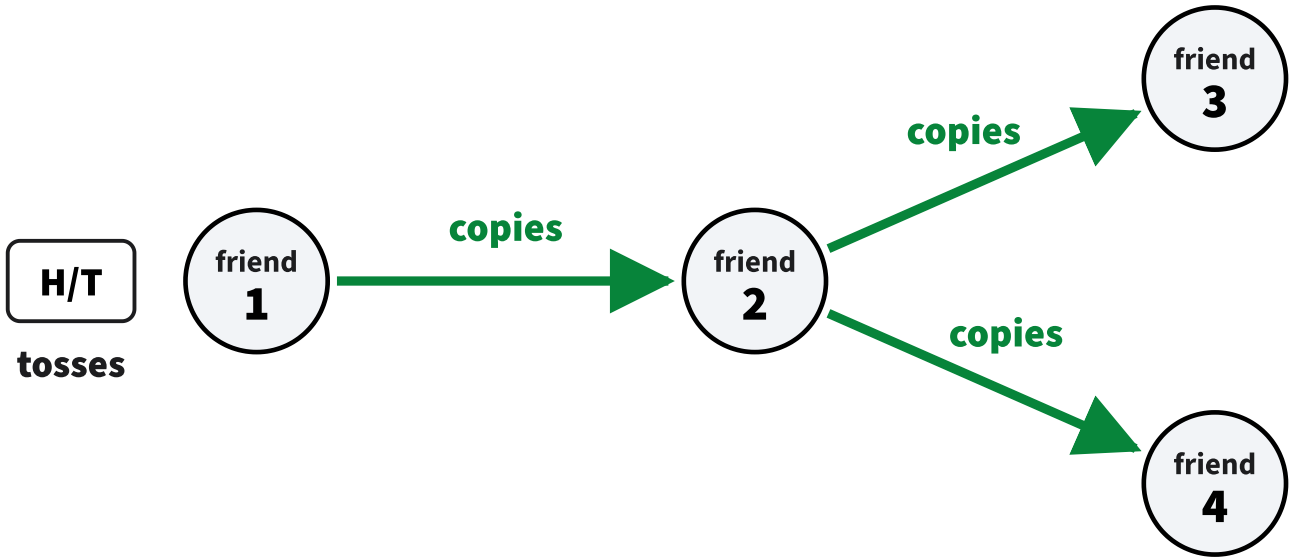
the information on its links | its number of links | its strongest link only

Answer. The information on its links: Theorem B, next.

Next: one number for a whole map.

A tree map says how every day is made

A start friend **tosses**. Along each link, the next friend usually writes the **same face**. So each of the $2 \cdot 2 \cdot 2 \cdot 2 = 16$ kinds of day gets a **chance**.



each copy is right most of the time; follow the links: every day gets a chance

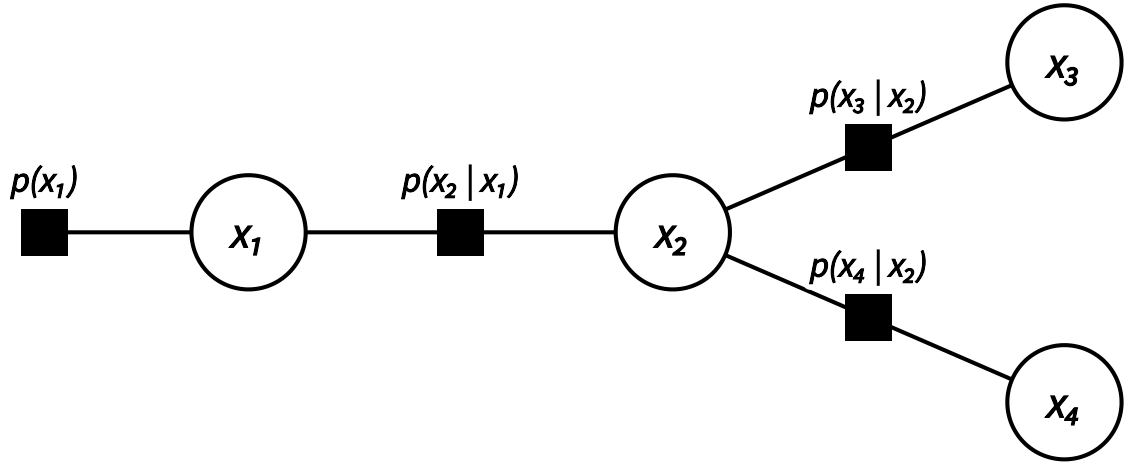
Each tree map is a model of the notebook.

A tree model: a start table times one table per link

Start at friend 1 and follow the links 1–2, 2–3, 2–4:

$$p_{\mathcal{T}}(\boldsymbol{x}) = \underbrace{p(x_1)}_{1\text{ tosses}} \cdot \underbrace{p(x_2 \mid x_1)}_{2\text{ given }1} \cdot \underbrace{p(x_3 \mid x_2)}_{3\text{ given }2} \cdot \underbrace{p(x_4 \mid x_2)}_{4\text{ given }2}$$

Without a start, any tree $\mathcal{T}$ (not loops; **(Lecture 7, p. 98)**): $p_{\mathcal{T}}(\boldsymbol{x}) = \prod_i p(x_i) \cdot \prod_{(i,j)\in\mathcal{T}} \frac{p(x_i, x_j)}{p(x_i) \cdot p(x_j)}$



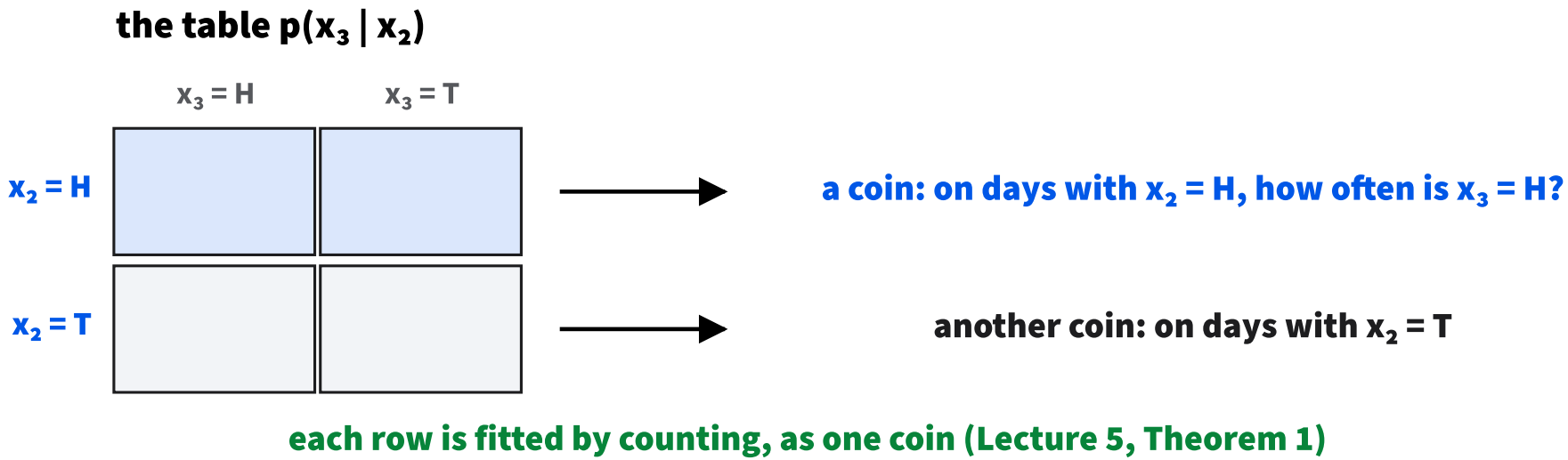
one square per table: a tree factor graph, so sum-product is exact on it

A tree model: a start table, one table per link.

Each table of a tree is fitted by counting

Each row of a link's table is **a coin**: when friend 2 shows H, how often does friend 3? Counting fits it **(Lecture 5, p. 11)**.

The tables that make the notebook most likely: $\hat{p}(x_j \mid x_i) = \dfrac{n(x_i, x_j)}{n(x_i)}$, counted fractions **(Lecture 7, p. 96)**

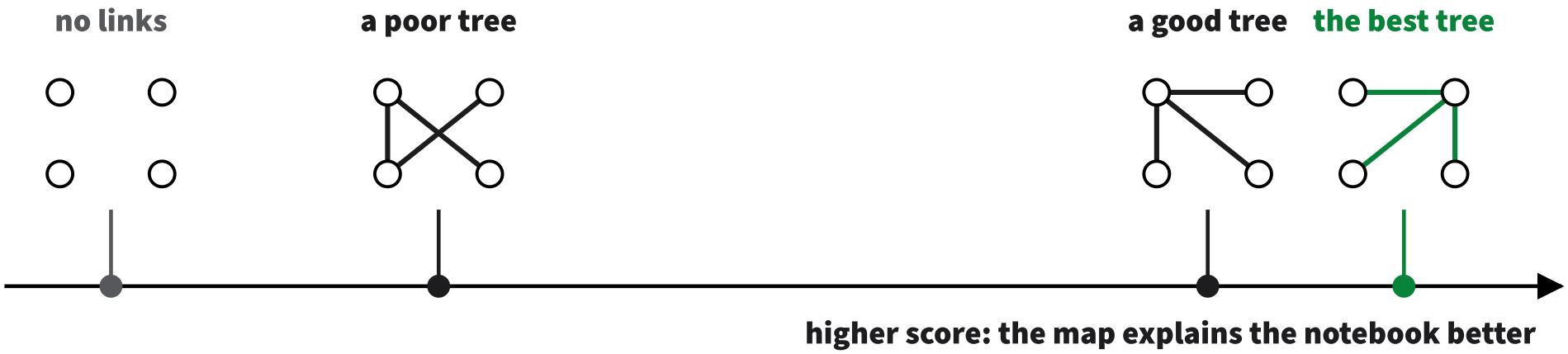


Fitting a tree needs no search: just count.

The score of a map: how likely it makes the notebook

A map's **score**: the notebook's log-likelihood per day, with counted tables $\hat{p}_{\mathcal{T}}$.

$$\ell(\mathcal{T}) = \frac{1}{N}\sum_{d=1}^{N} \ln \hat{p}_{\mathcal{T}}\left(\boldsymbol{x}^{(d)}\right), \text{ with } \boldsymbol{x}^{(d)} \text{ the faces on day } d$$



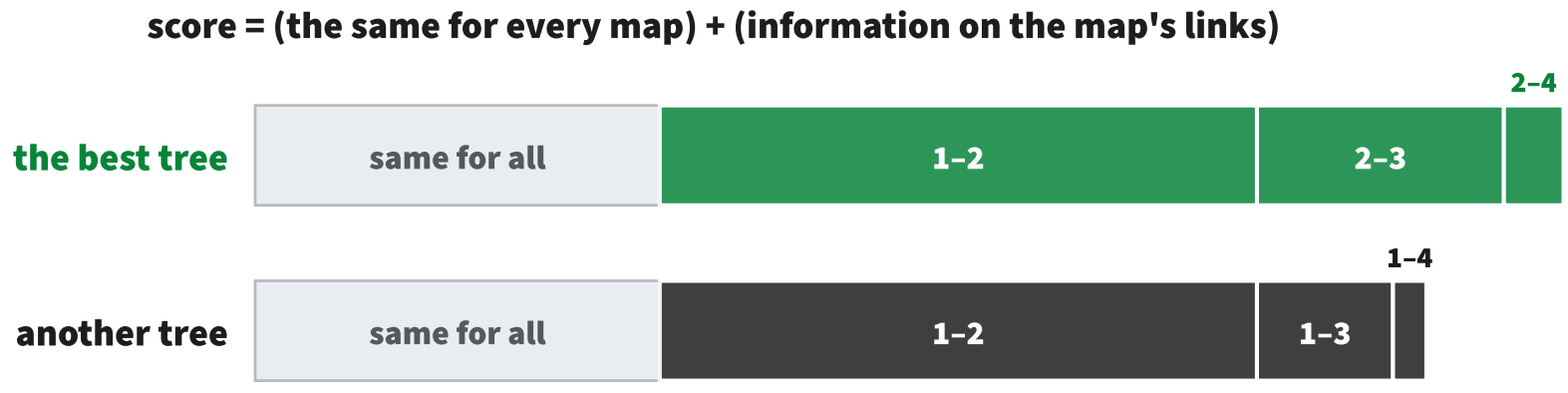
A higher score: the map explains the notebook better.

Theorem B: a tree's score is the information on its links

Theorem B (Chow and Liu). For every tree $\mathcal{T}$, with counted tables,

$$\ell(\mathcal{T}) = \sum_{(i,j)\in\mathcal{T}} \hat{I}(X_i; X_j) \; - \; \sum_{i} \hat{H}(X_i).$$

In words: information on the links, minus a part no map can change.



one block per link, as long as its information; the grey part (minus the entropies) is the same for every map

Insight. Compare maps by **adding their links' information**.

Remember: best tree = most information on its links.

Proof: the logarithm splits the score link by link

↳ Tree model without a start **(Lecture 7, p. 18)**; $\hat{p}(\boldsymbol{x}) = n(\boldsymbol{x})/N$, counted whole days.

$$\ell(\mathcal{T}) = \frac{1}{N}\sum_{d=1}^{N} \ln \hat{p}_{\mathcal{T}}\left(\boldsymbol{x}^{(d)}\right) = \sum_{\boldsymbol{x}} \hat{p}(\boldsymbol{x}) \cdot \ln \hat{p}_{\mathcal{T}}(\boldsymbol{x})$$

group equal days: $n(\boldsymbol{x})/N = \hat{p}(\boldsymbol{x})$

$$= \sum_{i}\sum_{\boldsymbol{x}} \hat{p}(\boldsymbol{x}) \cdot \ln \hat{p}(x_i) + \sum_{(i,j)\in\mathcal{T}}\sum_{\boldsymbol{x}} \hat{p}(\boldsymbol{x}) \cdot \ln \frac{\hat{p}(x_i, x_j)}{\hat{p}(x_i)\cdot\hat{p}(x_j)}$$

log of a product = sum of logs

$$= \sum_{i}\sum_{x_i} \hat{p}(x_i) \cdot \ln \hat{p}(x_i) + \sum_{(i,j)\in\mathcal{T}}\sum_{x_i,x_j} \hat{p}(x_i, x_j) \cdot \ln \frac{\hat{p}(x_i, x_j)}{\hat{p}(x_i)\cdot\hat{p}(x_j)}$$

sum out the unused faces

$$= -\sum_{i} \hat{H}(X_i) + \sum_{(i,j)\in\mathcal{T}} \hat{I}(X_i; X_j)$$

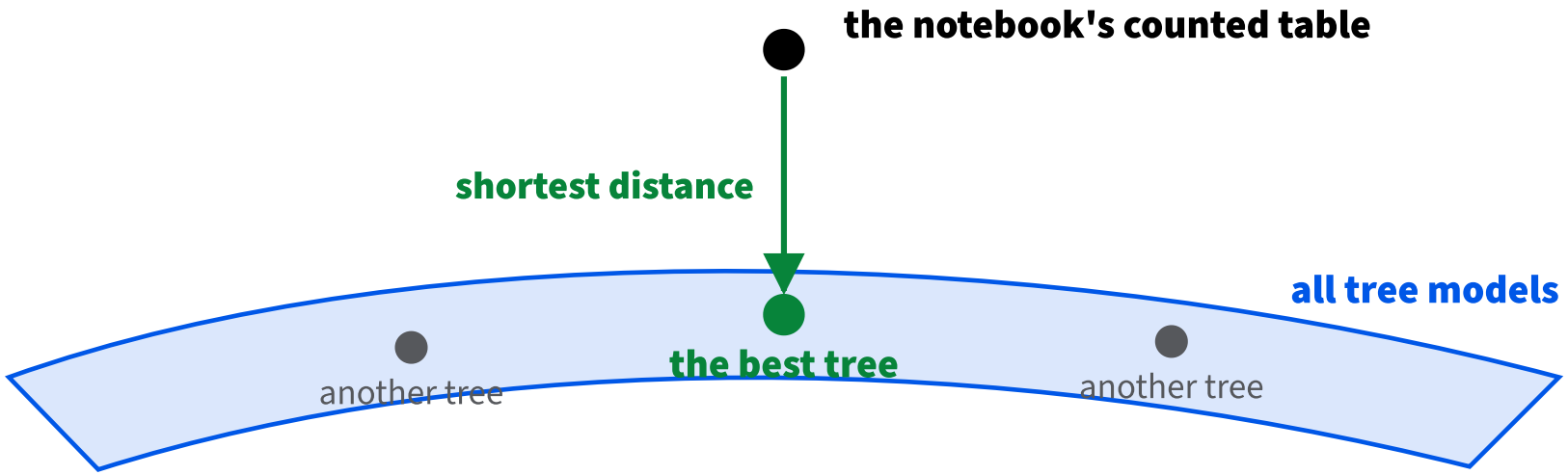
definitions of entropy and information

Each link adds its information; the rest is fixed.

The best tree is the tree closest to the notebook

$\ell(\mathcal{T}) = -\hat{H}(\boldsymbol{X}) - \mathrm{KL}\left(\hat{p} \,\|\, \hat{p}_{\mathcal{T}}\right)$ **(Lecture 7, p. 97)**, $\hat{H}(\boldsymbol{X})$: entropy of counted days

1. $\hat{H}(\boldsymbol{X})$ is the same for every map.
2. So the best tree **minimizes the distance** $\mathrm{KL}(\hat{p} \,\|\, \hat{p}_{\mathcal{T}})$, counted table first, as in Lecture 6 **(Lecture 6, p. 43)**.



Highest score = closest tree to the notebook.

Why trees: exact answers, exact search, few numbers

1. **Exact answers:** sum-product is exact on a tree **(Lecture 6, p. 26)**.
2. **Exact search:** the best tree is found quickly (next section).
3. **Few numbers:** each non-start friend copies one parent.

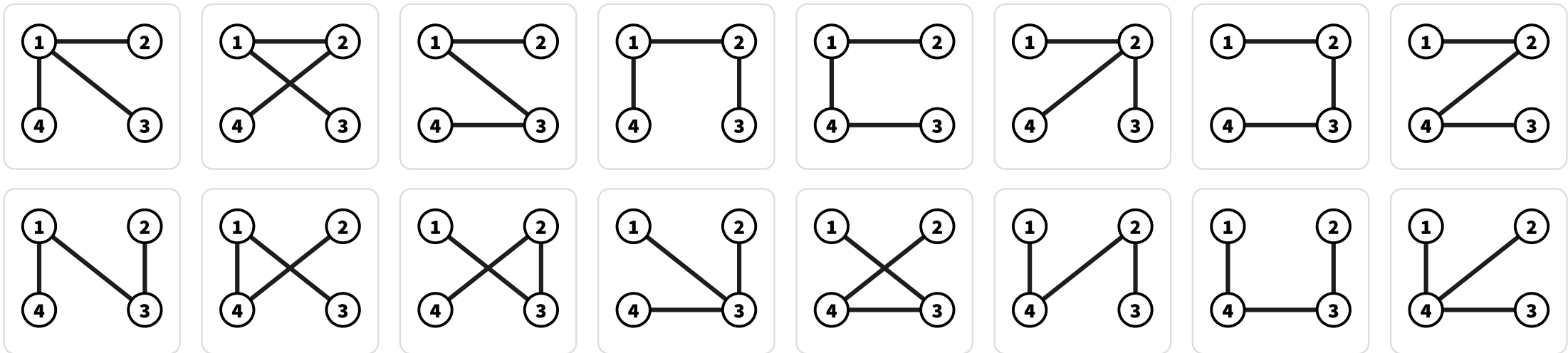
what is fitted	count	4 friends	30 friends
start friend: chance of H	**1**	**1**	**1**
each other friend: chance of H given parent H, given parent T	$2 \cdot (m-1)$	**6**	**58**
a tree, in total	$2m-1$	**7**	**59**
full table: 2^m chances, adding to 1	2^m-1	**15**	**over a billion**

The price: a tree cannot hold a loop.

Trees: exact, fast, small; loops are the price.

Next: which tree carries the most information?

Among **all trees**, find the **largest total information**.



every way to link four friends with no loop: $4^2 = 16$ trees

Four friends have 16 trees. About how many trees do thirty friends have?

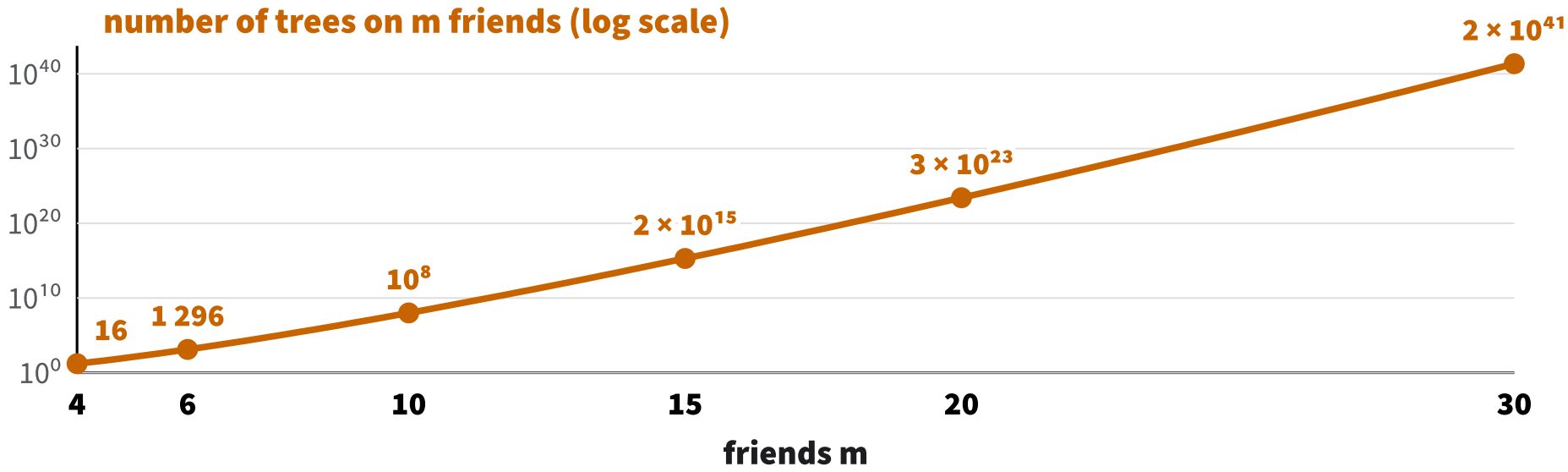
about 900 | about 30 000 | about 10^{41}

Answer. About 10^{41}: 30^{28} trees, by Cayley's formula, next.

Trying every tree is hopeless; we need a rule.

The number of trees explodes with the number of friends

Cayley's formula: m friends can be linked into m^{m-2} different trees.

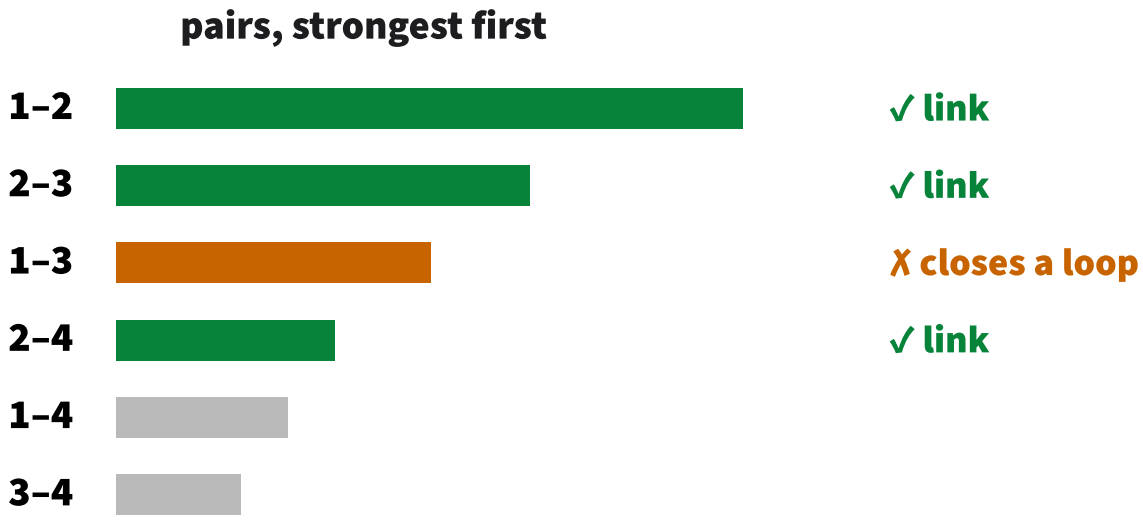


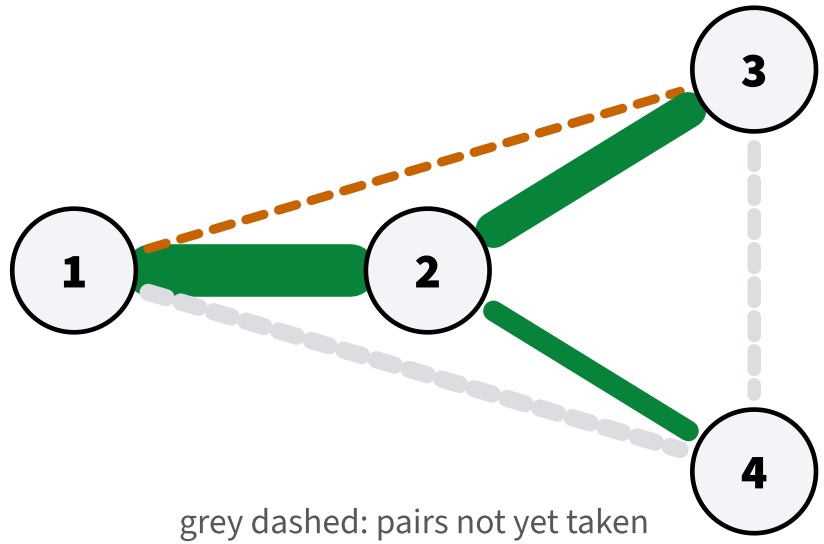
4 friends: $4^2 = 16$ trees. 30 friends: $30^{28} \approx 2 \times 10^{41}$ trees.

Listing trees works for four friends, never thirty.

Greedy rule: strongest link first, but never a loop

1. **Sort** the pairs by their information, strongest first.
2. **Take** the next pair, **unless it closes a loop**.
3. **Stop** at $m - 1$ links: everyone is connected.





Each step is easy: compare, then check for a loop.

Kruskal's algorithm finds a maximum spanning tree

Definition. **Spanning tree**: $m - 1$ links joining all m friends, no loop. **Maximum spanning tree**: largest total weight (here, information). **Kruskal's algorithm**: the greedy rule.

Chow–Liu tree (definition): $\hat{\mathcal{T}} = \arg\max_{\mathcal{T}} \sum_{(i,j)\in\mathcal{T}} \hat{I}(X_i; X_j)$, found by Kruskal



count, measure, sort, link: no search over trees

Chow–Liu = count, measure, then Kruskal.

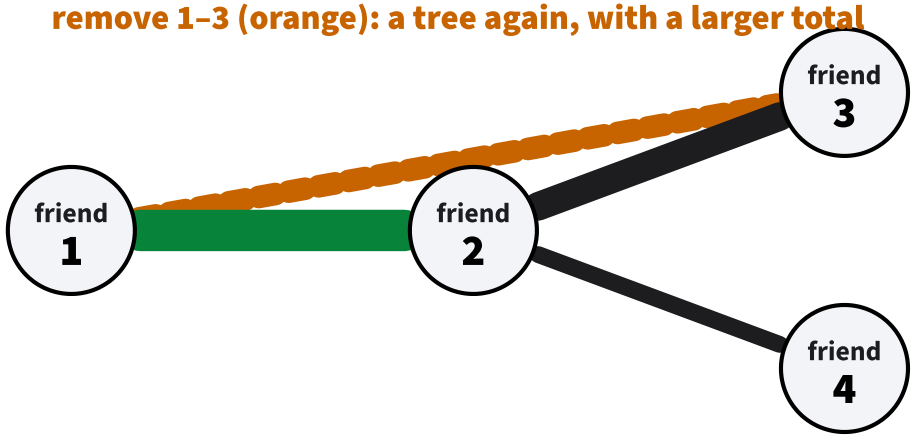
Theorem C: the greedy rule always finds the best tree

Theorem C (Kruskal). The greedy rule returns a spanning tree with the **largest** total information of all spanning trees.

In words: the strongest link always fits in some best tree.

Why: take a tree **without** the strongest link 1–2.

1. **Add 1–2.** This makes one loop: 1–2–3.
2. **Remove another loop link**, 1–3: again a tree.
3. 1–2 is strongest, so the swap **never lowers the total**. Same at every step **(Lecture 7, p. 30)**.

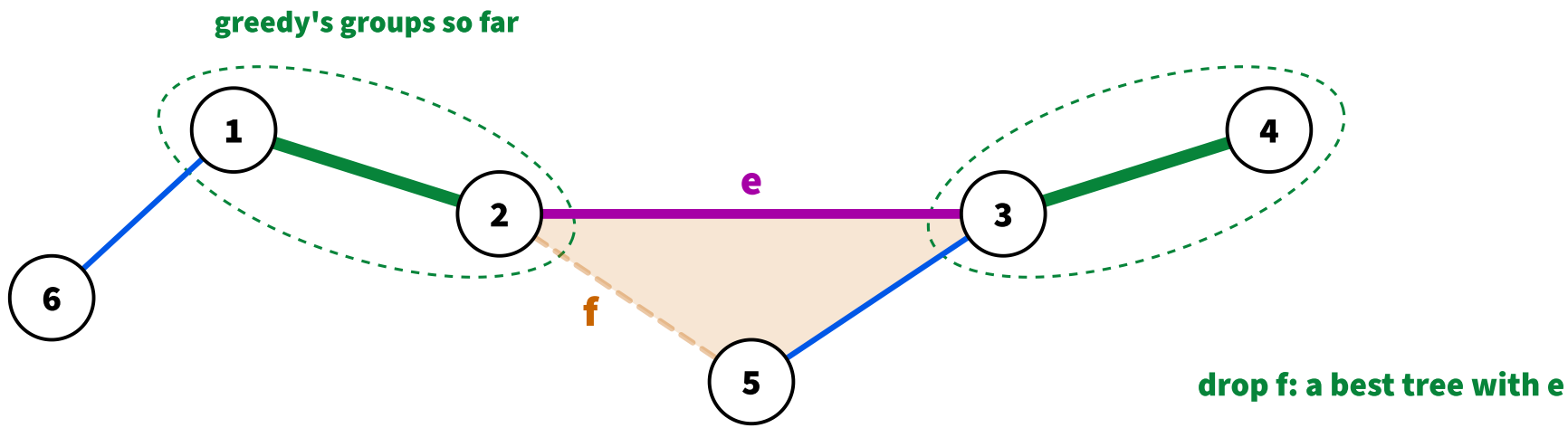


The strongest safe link costs nothing: greedy is exact.

Proof: swapping in the greedy link never lowers the total

↳ **Claim:** after every greedy step, some best tree $\mathcal{T}$ (blue) holds all links greedy has taken (green). True at the start.

1. Greedy takes e (magenta). If $e \in \mathcal{T}$, done. Else:
2. $\mathcal{T} + e$ has one loop (shaded).
3. The loop crosses greedy's groups at a blue link f (orange) not yet taken. f was allowed, greedy chose e: $w(e) \geq w(f)$.
4. Swap: $w(\mathcal{T} + e - f) = w(\mathcal{T}) + w(e) - w(f) \geq w(\mathcal{T})$. A best tree holding e.



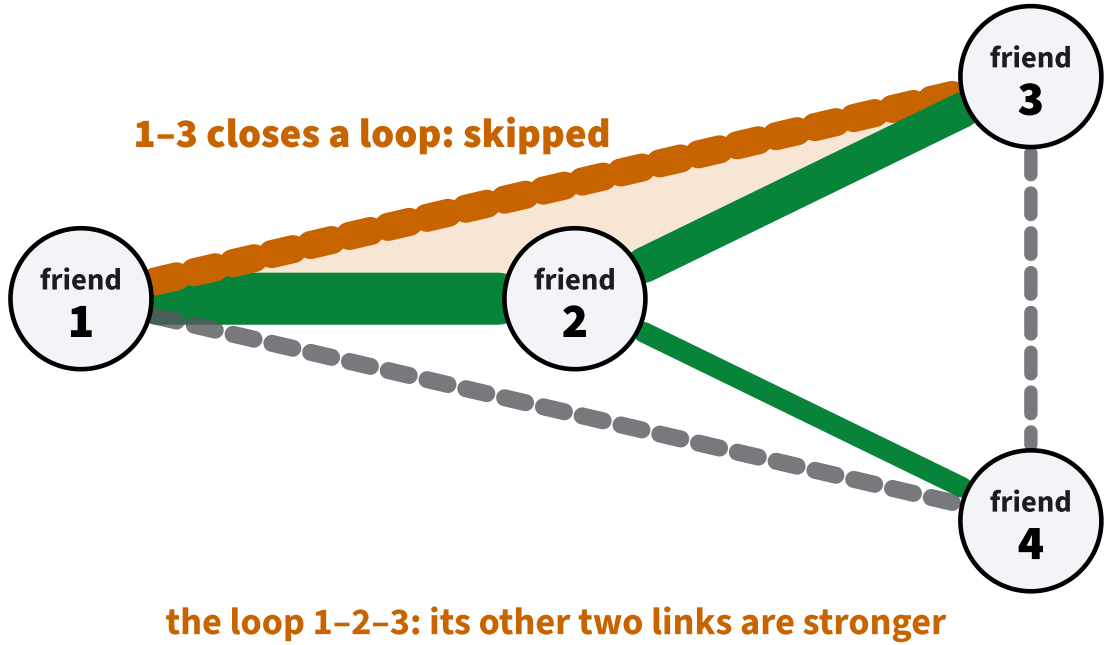
Swap by swap, a best tree becomes greedy's tree.

With exact information, greedy finds the true links

Question: with an endless notebook (exact information), does greedy find the true links?

1. **An indirect pair is weaker than its path.** 1–3 is weaker than 1–2 and 2–3 (Theorem A).
2. **So greedy meets 1–2 and 2–3 first.** Then 1–3 would close a loop: skipped. Same for every indirect pair.

(Needs each copy chance between ½ and 1.)



Exact information gives the true tree; real notebooks are next.

The greedy rule needs only pair counts, not a search

$m(m-1)/2$ pairs, counted in **one pass** over N days: about $m^2 N/2$ counts, plus a sort.

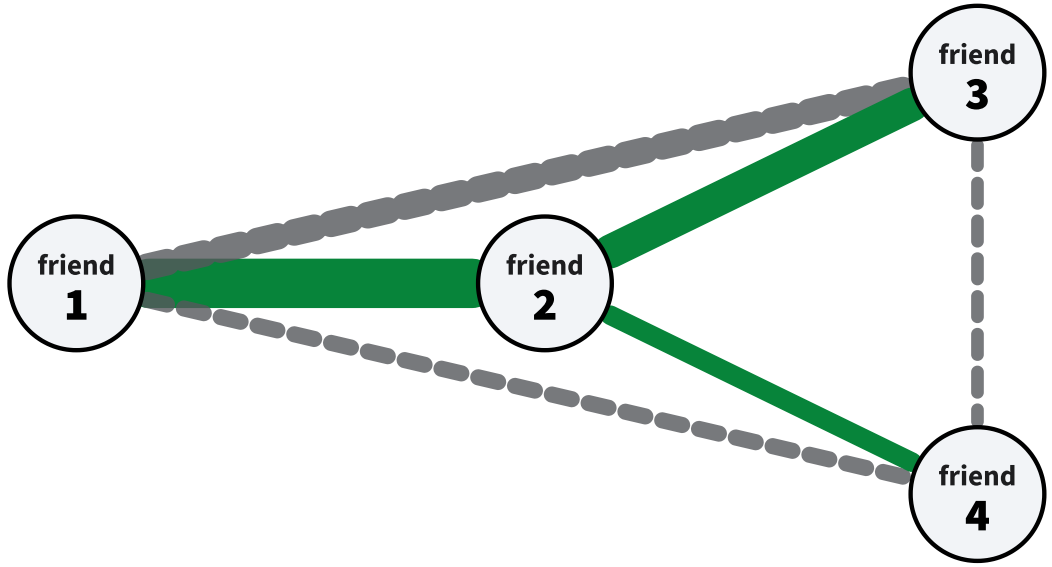
friends m	trees to list	pairs to count
4	**16**	**6**
30	2×10^{41}	**435**
1000	10^{2994}	**499 500**

1000 friends: half a million pair tables.

Counting pairs grows like m^2; listing trees grows faster than any power.

Next: a real notebook is short; can we trust its map?

Section 3 assumed **exact** information. Few days make each count **a little off**.



green: true links · grey dashed: pairs linked only through others · thickness: strength

Two friends toss on their own, so their true information is 0. Counted from only 20 days, their information is …

exactly zero | above zero | below zero

Answer. Above zero, almost always: agreeing on 12 of 20 days by chance looks like a weak link.

Next: how far off can a short notebook be?

A short notebook makes unlinked friends look linked

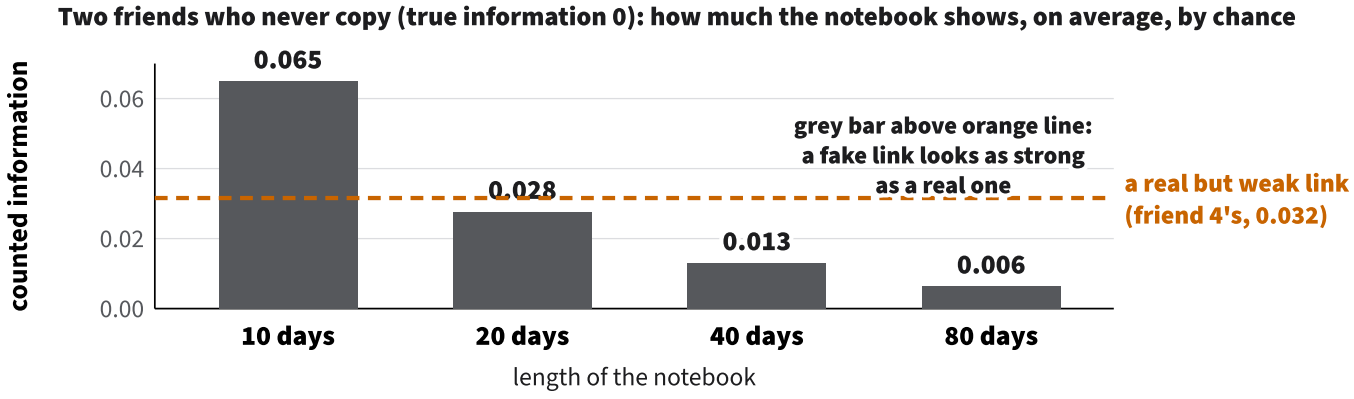
PROBLEM A real notebook is short. Friends who never copy still agree on some days, **by chance**.

WE DO Count the information of a pair that never copies, in many 20-day notebooks.

WHY Greedy takes the largest counts; a chance count can push out a true link.

Guarantee (Wilks). If friends i and j are independent, then for large N, $2N \cdot \hat{I}(X_i; X_j)$ behaves like a chi-square variable with 1 degree of freedom, whose average is 1.

In words: chance information averages about $1/(2N)$.



SO **Chance alone gives about $1/(2N)$ (Lecture 7, p. 99); with 20 days, a fake pair matches friend 4's real link in 29% of notebooks.**

NEXT Do these mistakes disappear when the notebook grows?

Theorem D: with enough days, the tree is right

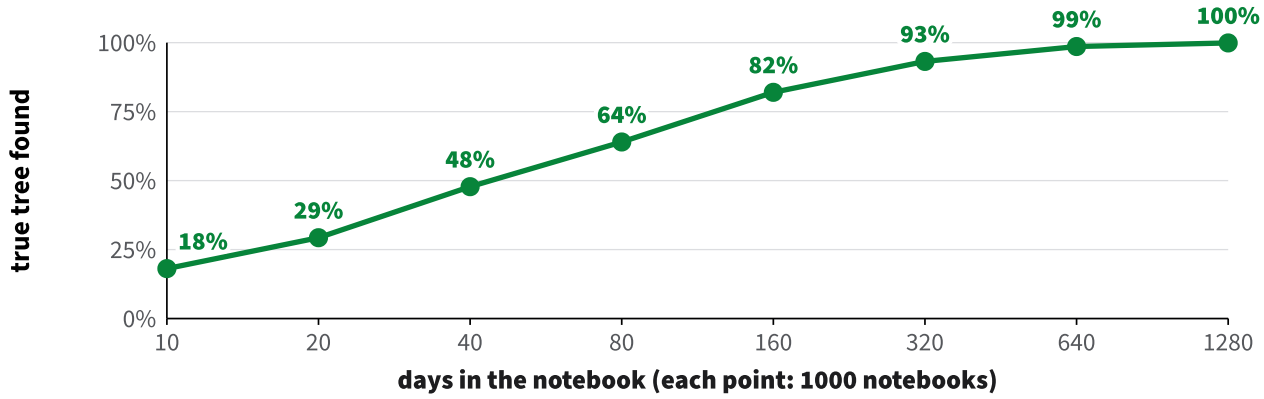
PROBLEM Chance counts shrink like $1/(2N)$, but never reach 0.

WE DO Ask how fast the chance of a wrong tree falls as days grow.

WHY It tells us how many days a trustworthy map needs.

Theorem D (consistency). If the friends copy along a tree (each copy chance between ½ and 1, friend 1's coin not certain), the chance that the Chow–Liu tree (Lecture 7, p. 28) is **wrong** shrinks to 0 like e^{-cN}, some $c > 0$, as the days N grow (proofs: references).

In words: more days, fewer wrong maps.



SO **Our friends: right in 29% of notebooks with 20 days, 93% with 320, 99% with 640.**

NEXT In a short notebook, which link goes wrong?

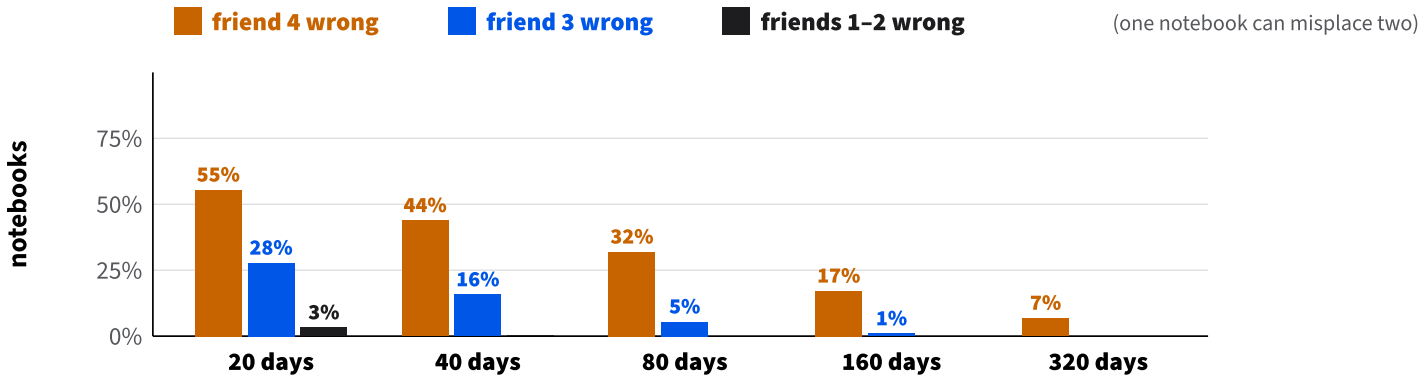
Most mistakes misplace friend 4, the weakest link

PROBLEM A wrong tree tells us nothing until we know **where** it is wrong.

WE DO Sort the wrong trees of 1000 notebooks per length by the friend misplaced.

WHY The link that fails most is the one to check first.

↳ **True tree:** 2 copies 1 (7/8); 3 and 4 copy 2 (3/4, 5/8) **(Lecture 7, p. 59)**.



Guarantee (Tan, Anandkumar, Tong, Willsky). The rate c of Theorem D is set by the hardest swap: a true link against the unlinked pair on its path closest to it in information.

In words: the weakest link decides how many days we need.

SO **Friend 4 copies with only 5/8, so its weak link nearly ties with the indirect pair 1–4.**

NEXT Can we refuse a link that is too weak for the days we have?

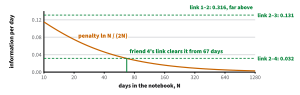
Keep a link only if it beats what chance can produce

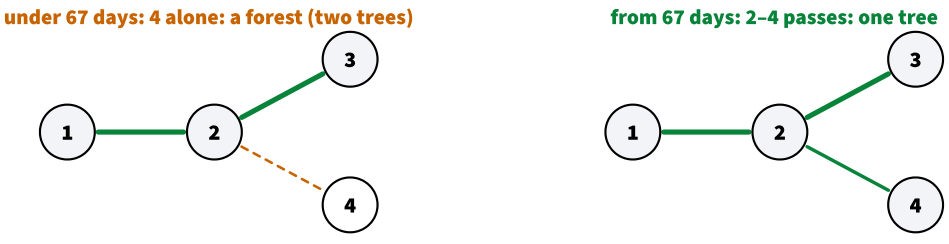
PROBLEM Greedy always adds 3 links, even when a link is only chance.

WE DO Run greedy, but **skip pairs below a threshold** set above the chance level $1/(2N)$ **(Lecture 7, p. 34)**.

WHY A link the days cannot support does more harm than no link.

> keep the link i–j only if $\hat{I}(X_i; X_j) > \dfrac{\ln N}{2N}$





Guarantee (Tan, Anandkumar, Willsky). If the threshold $t_N \to 0$ but $N \cdot t_N \to \infty$, as for $t_N = \ln N/(2N)$, the chance that the learned forest is wrong goes to 0.

In words: fake links fall away, true links stay.

SO **The threshold is the Bayesian information criterion (Lecture 7, p. 100); skipped pairs leave a forest.**

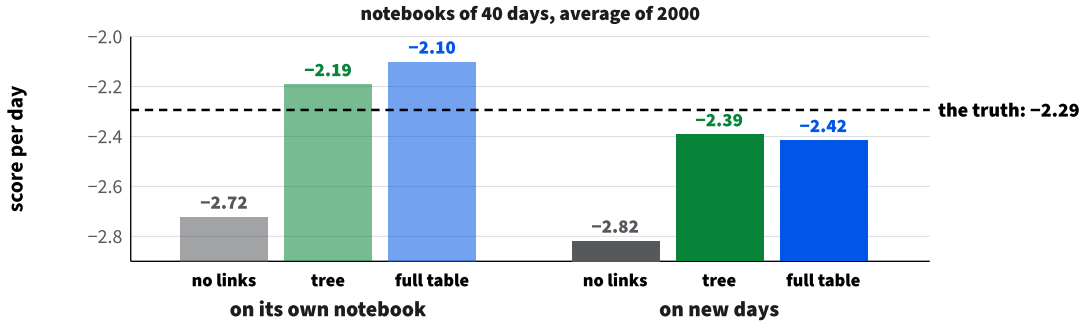
NEXT Why not keep every link? More links score higher.

More links always raise the notebook's own score

PROBLEM Each link adds information, **never negative** (Theorem B); the **full table** $\hat{p}$ scores highest **(Lecture 7, p. 97)**.

WE DO Score each model on **new days** it never saw.

WHY A good map predicts new days, not only the days it learned from.



score on new days $= \frac{1}{M} \cdot \sum_{m=1}^{M} \ln q(\text{new day } m)$: higher is better

q: the model learned from the 40 days; M: new days.

Guarantee (Akaike). As M grows, the score tends to $-H(p) - D(p \,\|\, q)$. If the model can express p and has k free numbers, the average of $D(p \,\|\, q)$ is about $k/(2N)$ for large N: full table $k = 15$, tree $k = 7$.

In words: extra numbers cost about 1/(2N) each on new days.

SO **2000 simulated notebooks of 40 days (Lecture 7, p. 104): tree −2.39 beats full table −2.42.**

NEXT One notebook gives one map. How sure is each link?

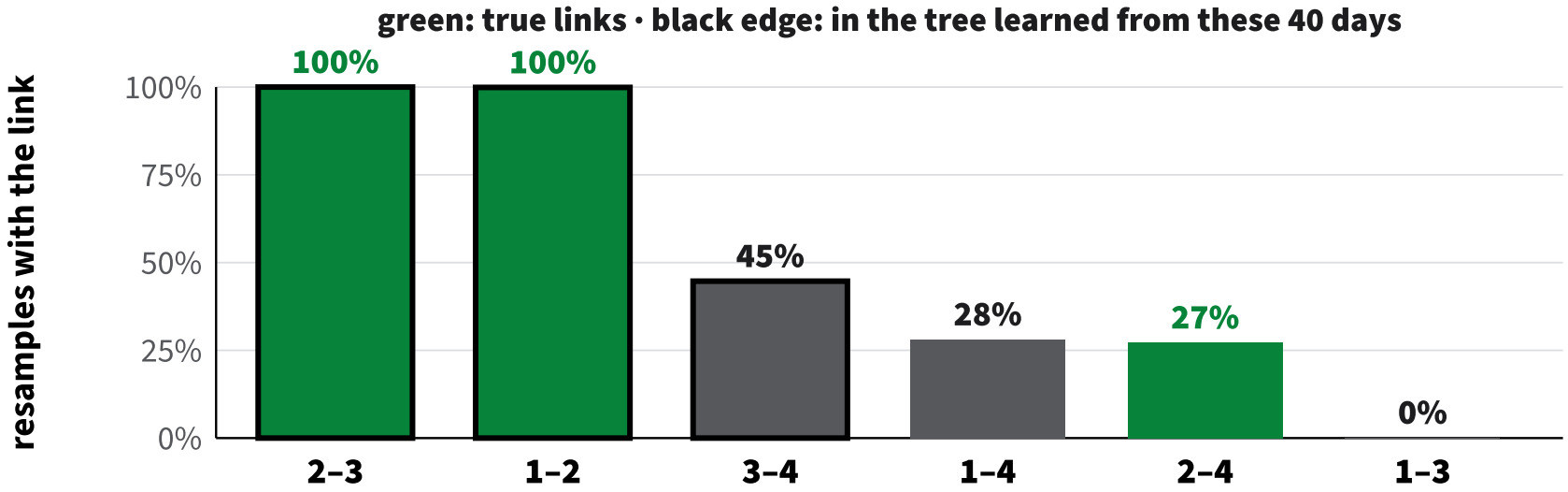
Resampling the notebook shows which links are stable

PROBLEM Only **one** 40-day notebook.

WE DO Pick 40 of our days **with replacement**, run greedy; **repeat 1000 times**.

WHY Links resting on **lucky days** come and go; strong links **stay**.

↳ **With replacement:** each pick is **uniform** (every day 1/40) and **independent**; a day may repeat. This is the **bootstrap**.



SO **1–2 and 2–3: 999 and 1000 of 1000. Friend 4 agrees with friends 2, 3, 1 on 23, 23, 22 days: a near-tie, so it splits: friend 3 in 45%, friend 1 in 28%, friend 2 in 27%.**

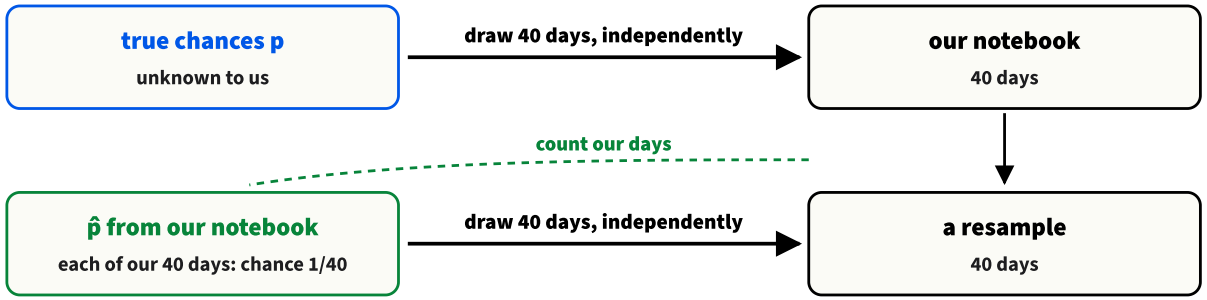
NEXT Why may resamples stand in for **new real notebooks**?

Resampling our days imitates collecting new ones

PROBLEM We read resamples as if they were **new notebooks**.

WE DO Compare the **two recipes**.

WHY Same recipe ⇒ **trustworthy shares**.



same recipe, p̂ in place of p; as N grows p̂ → p, so resamples behave like new real notebooks

Guarantee (Efron).

1. **Same recipe:** N independent draws, from $\hat{p}$ instead of p.
2. $\hat{p} \to p$ as N grows (law of large numbers).

In words: so a link's share of resamples ≈ its chance in a new notebook.

SO **1–2 beats its rival 1–3 clearly (35 vs 29 agreeing days): 999 of 1000. Friend 4's pairs nearly tie (23, 23, 22): these days cannot decide.**

NEXT A link is stable. But **who copies whom**?

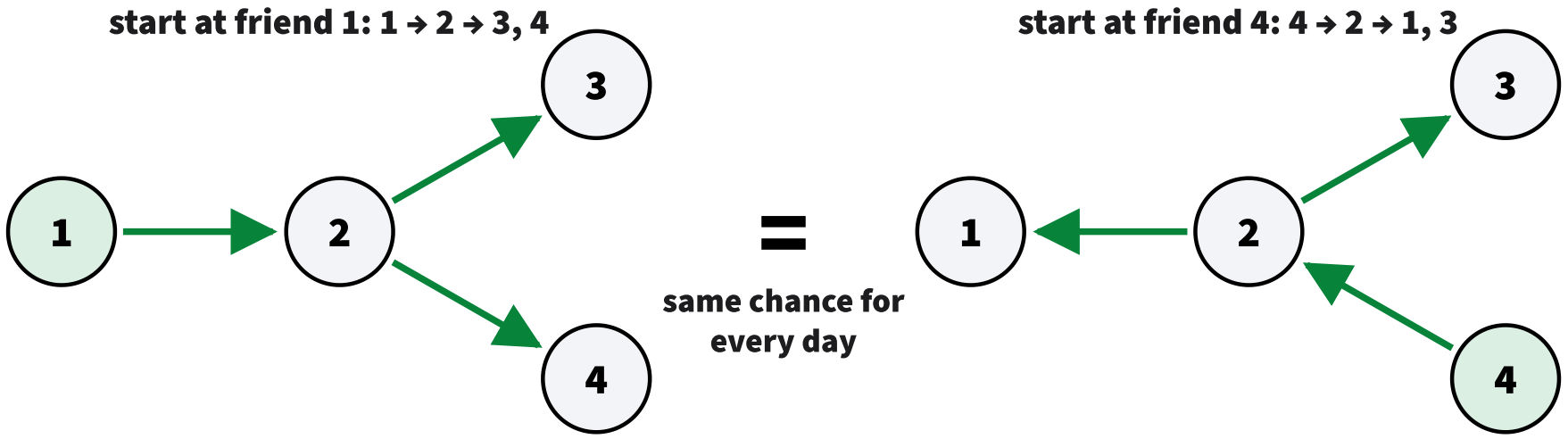
The notebook shows who is linked, not who copies whom

PROBLEM A link says two friends are tied, not which one copies.

WE DO Write the same tree starting from friend 1 and from friend 4.

WHY If both give every day the same chance, no notebook can tell them apart.

$$p(x_1) \cdot p(x_2 \mid x_1) = p(x_1, x_2) = p(x_2) \cdot p(x_1 \mid x_2)$$ (Lecture 7, p. 98)



to find an arrow, experiment: force friend 2's face; whoever then changes copies friend 2

SO **Information is not causation** (Lecture 3, p. 72)**: arrows need experiments or assumptions.**

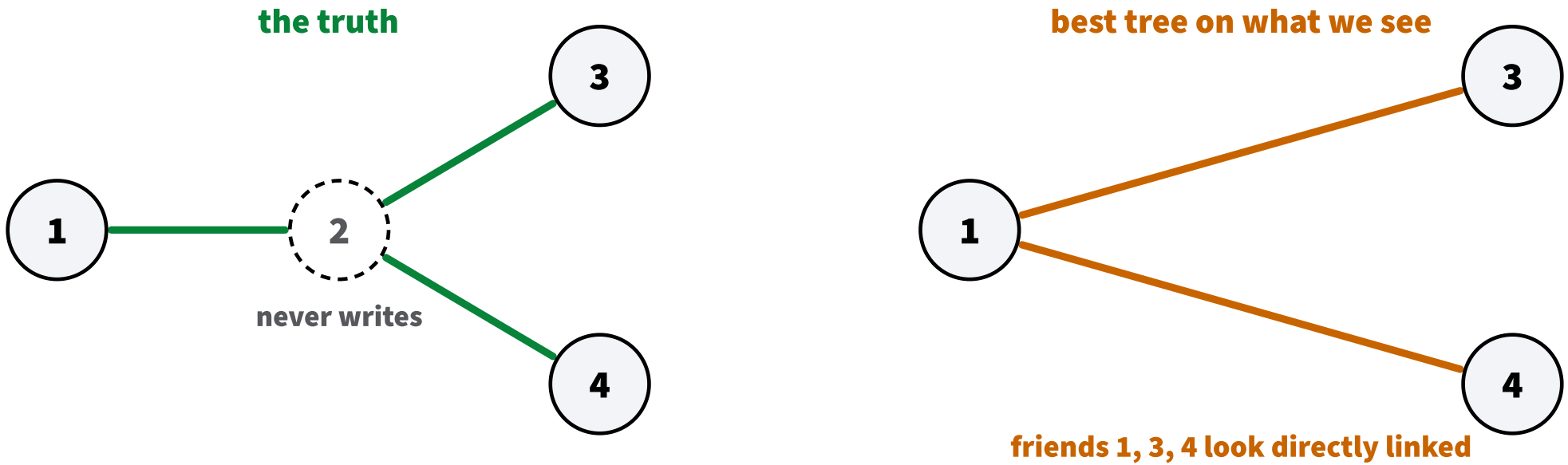
NEXT What if a friend never writes in the notebook?

A hidden friend makes the others look directly linked

PROBLEM Real notebooks rarely record everyone who matters.

WE DO Hide friend 2's column and learn from friends 1, 3, 4 **(Lecture 7, p. 101)**.

WHY Friends 1, 3, 4 agree only **through** friend 2. Hide friend 2, and nothing left in the notebook explains their agreement.



SO **1, 3, 4 look directly linked. Fix: latent tree models, tables fitted by Lecture 5's expectation–maximization (EM).**

NEXT Can the notebook still **reveal** the hidden friend?

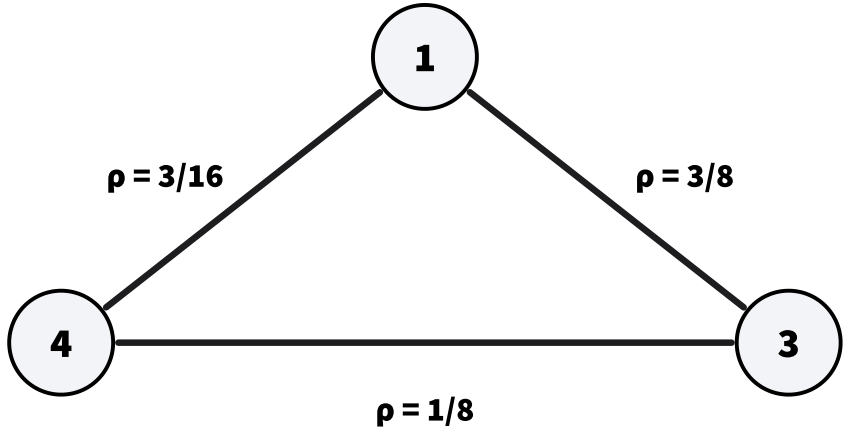
A hidden friend leaves a fingerprint in the agreements

PROBLEM Friend 2 **never writes**; we see only 1, 3, 4.

WE DO Turn each pair's agreement into a **strength**.

WHY Along a copying path, strengths **multiply**.

Main idea: a strength ρ runs from 0 (no tie) to 1 (always the same face).
Why they multiply: a link with strength ρ acts like this: with chance ρ it **passes the face on**; otherwise it writes a fresh fair toss. A face crosses two links only if **both pass it on**: chance $\rho_{12} \cdot \rho_{23}$ **(Lecture 7, p. 101)**.



SO **Try friend 1 as the centre: then 3 and 4 should agree less than they do. Friends 3 and 4 fail the same way. So the centre is someone we cannot see.**

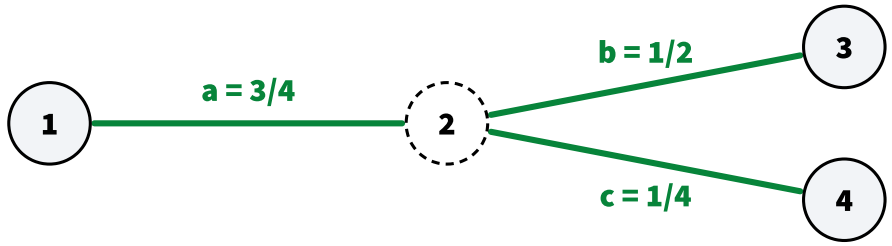
NEXT Can we compute the hidden centre's links?

Three seen strengths give the hidden friend's three links

PROBLEM The centre's links a, b, c are **unknown**.

WE DO Each seen pair is a path through the centre: **three equations**.

WHY Three equations fix three unknowns.



$$a \cdot b = \tfrac{3}{8}, \quad a \cdot c = \tfrac{3}{16}, \quad b \cdot c = \tfrac{1}{8}$$

paths 1–3, 1–4, 3–4

$$a^2 = \frac{(a \cdot b) \cdot (a \cdot c)}{b \cdot c} = \frac{3/8 \cdot 3/16}{1/8} = \tfrac{9}{16} \Rightarrow a = \tfrac{3}{4}$$

first × second ÷ third

$$b = \tfrac{3}{8} \div \tfrac{3}{4} = \tfrac{1}{2}, \quad c = \tfrac{3}{16} \div \tfrac{3}{4} = \tfrac{1}{4}$$

divide by a

Guarantee (Choi, Tan, Anandkumar, Willsky). If each hidden friend links **at least three** others, no strength 0 or ±1, the seen pairs **determine the hidden tree**.

SO **Undo the strength ($\rho = 2 \cdot P - 1$ gives $P = (1 + \rho)/2$): a, b, c become copy chances 7/8, 3/4, 5/8, friend 2's habits, recovered.**

NEXT Collect the checks before believing a map.

Before believing a map, make three checks

1. **Is each link above its penalty?** If not, drop it.
2. **Does each link survive resampling?** Report how often it appears.
3. **Does the map beat simpler maps on new days?**

And remember: **links are not arrows**, and **a hidden friend fakes links**.

Trust a map only after these checks.

Next: what if the true map has a loop?

So far: copying along a **tree**. Real friends may watch **two**.



from the notebook alone, to a map of links we can trust

Friend 4 copies friend 2 on some days and friend 3 on others; friend 3 copies friend 2. How many links does the true map have?

three, like a tree | four: one loop | six: every pair

Answer. Four: 1–2, 2–3, 2–4, 3–4, with the loop 2–3–4.

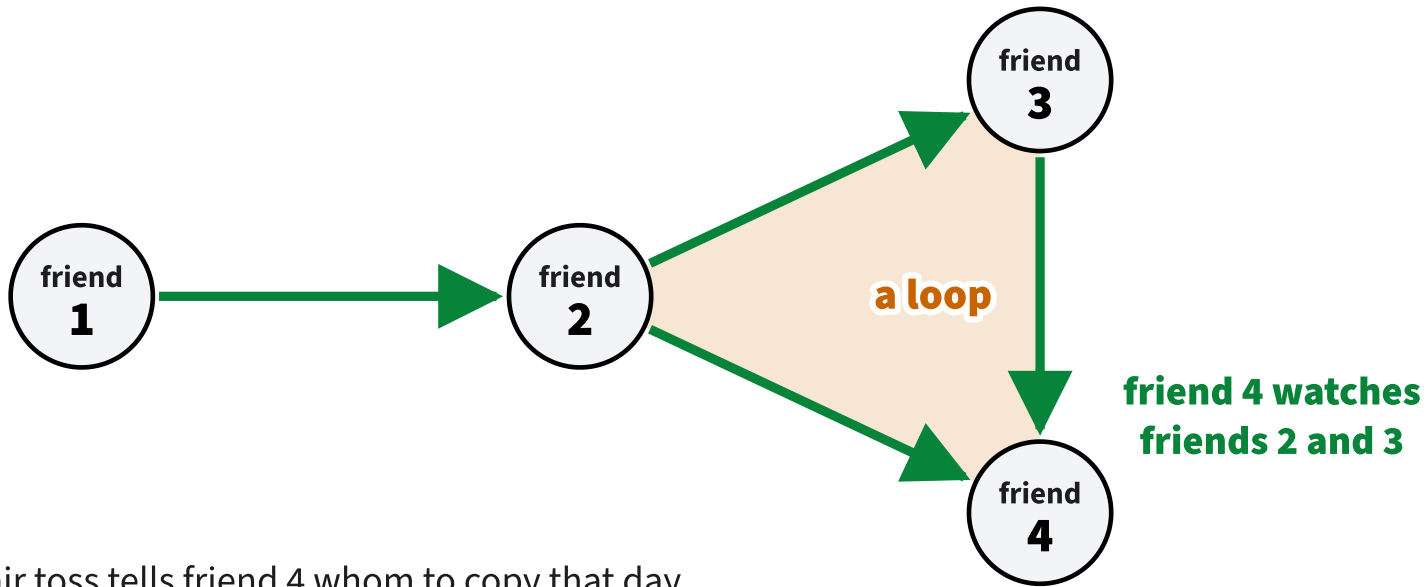
Next: maps with loops, beyond trees.

What if friend 4 watches two friends?

PROBLEM Real friends may copy more than one friend; a tree cannot show that.

WE DO Each day a fair toss picks whom friend 4 copies: friend 2 or friend 3.

WHY It is the smallest change that puts a loop into the truth.



a fair toss tells friend 4 whom to copy that day

SO **Friends 2, 3, 4 now form a loop: four true links, not three.**

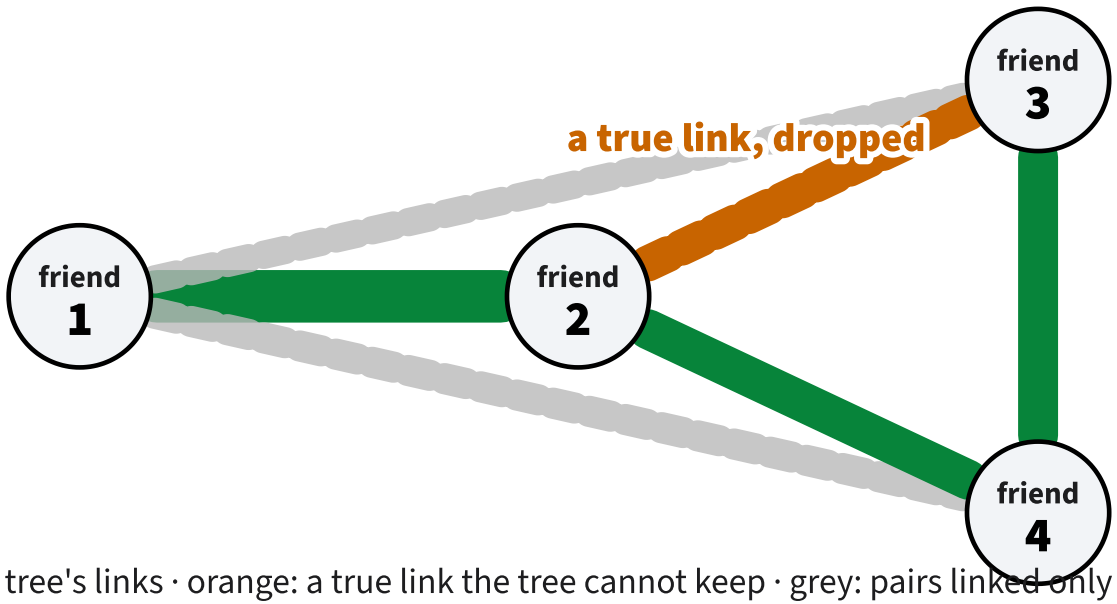
NEXT What does the tree method do with a loop?

The best tree must drop a true link of the loop

PROBLEM A tree on four friends has exactly 3 links; the truth has 4.

WE DO Run greedy on a perfect notebook of the loop **(Lecture 7, p. 76)**.

WHY To see which true link a tree is forced to lose.



green: the best tree's links · orange: a true link the tree cannot keep · grey: pairs linked only through others

SO **Greedy keeps 1–2, 2–4, 3–4 and drops the true link 2–3.**

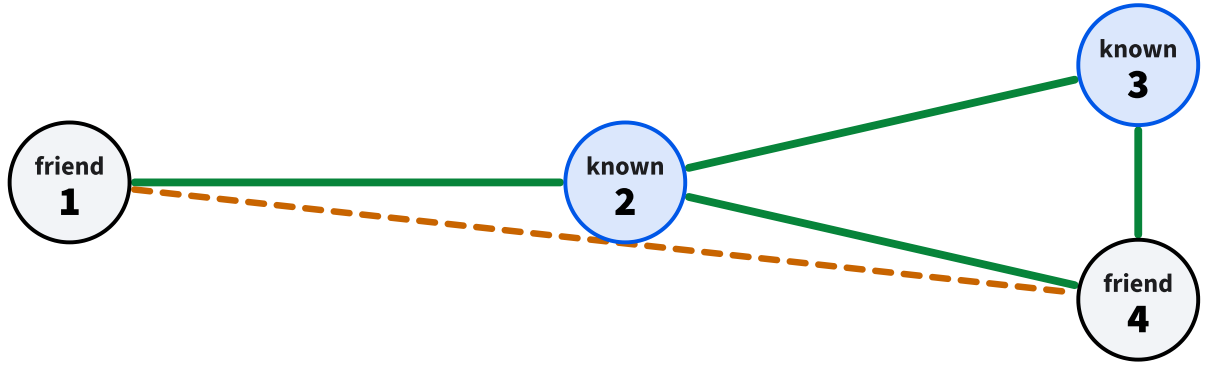
NEXT We need a test for one link that works without a tree.

No link means no information once the others are known

PROBLEM Friends 1 and 4 agree often, yet are **not linked**.

WE DO Ask: once friends 2 and 3 are **known**, does friend 1 still help guess friend 4?

WHY Friend 4 copies only 2 or 3, so friend 1 reaches 4 **only through them**.



1–4: with 2 and 3 known, friend 1 adds **nothing**: no link. **2–4:** even with 1 and 3 known, friend 2 still helps (4 copies it on half the days): link.

Theorem E (pairwise Markov property). If every day has positive chance, i and j are linked in the smallest fitting map **exactly when** X_i and X_j are dependent **given all the other friends** (Lecture 7, p. 105).

In words: a link is a tie the other friends cannot explain.

SO **Test a link given the other friends, not alone.**

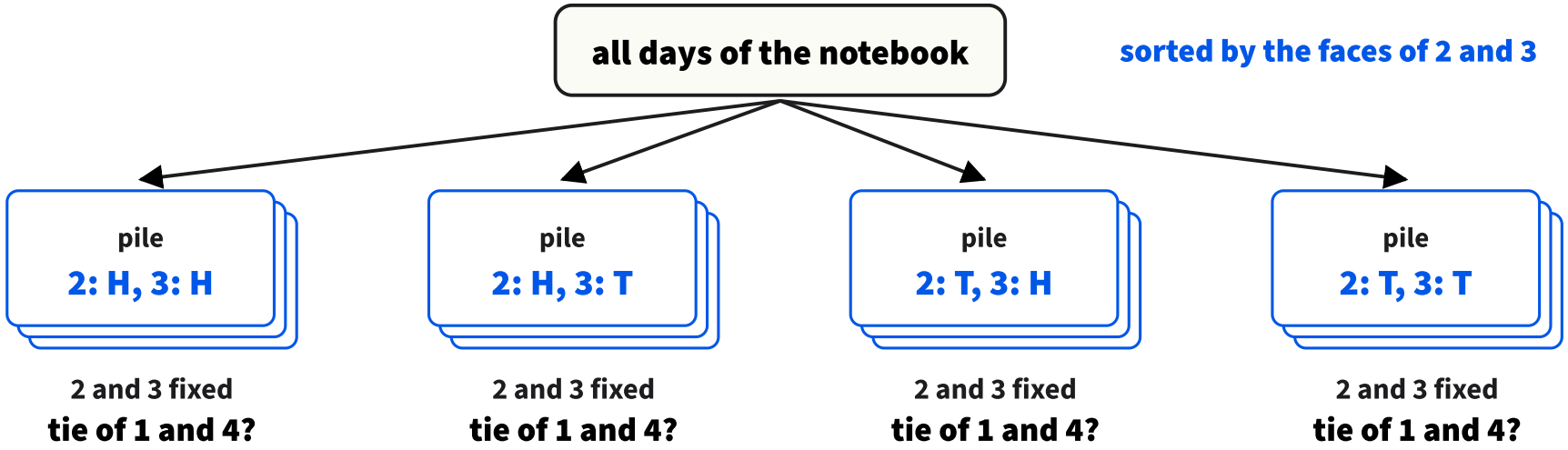
NEXT How do we measure information given other friends?

Conditional information measures a link given the others

PROBLEM Plain information of 1 and 4 also counts their tie **through** 2 and 3.

WE DO Sort days into **piles** by the faces of 2 and 3; measure 1–4 **inside each pile**; average.

WHY Inside a pile, 2 and 3 never change, so they cannot carry a tie: what is left is **direct**.



$$I(X_1; X_4 \mid X_2, X_3) = \sum_{\text{piles}} (\text{share of days}) \cdot (\text{information of } 1,\, 4 \text{ inside the pile})$$

SO **It is zero exactly when no pile shows a tie: 1 and 4 need no link. Any friends S work the same way.**

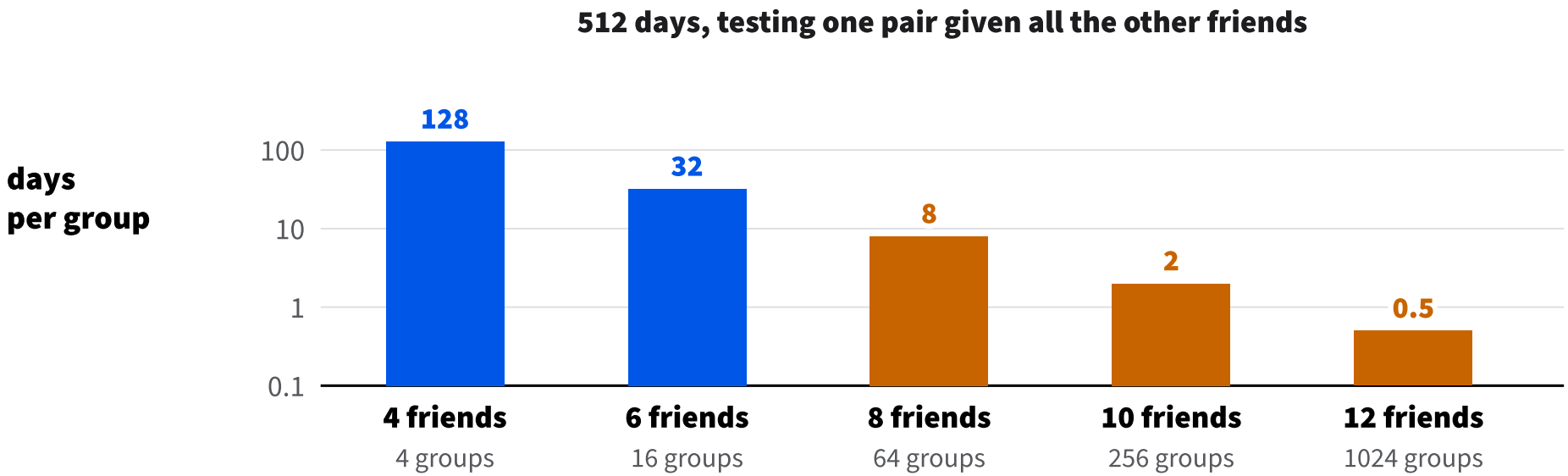
NEXT What does sorting into piles cost?

Conditioning on many friends needs many days

PROBLEM Given all $m - 2$ other friends, the days split into 2^{m-2} groups.

WE DO Count how many days each group keeps.

WHY A group with few days gives large chance counts again (Lecture 7, p. 34).



SO **Wilks again, with one degree of freedom per group: chance gives about (groups) ÷ (2N) (Lecture 7, p. 106): condition on as few friends as possible.**

NEXT A method that tests with small groups first.

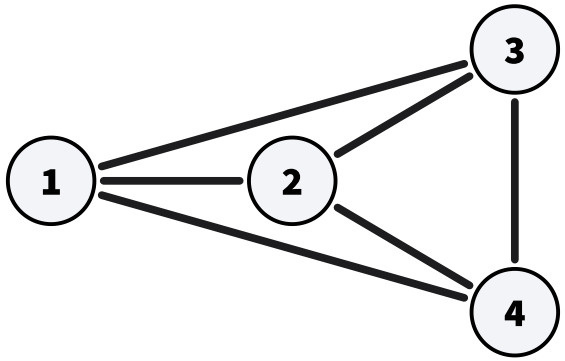
Test each link, and remove it when a few friends explain it

PROBLEM Testing given **all** others needs too many days.

WE DO The **PC algorithm** (for **P**eter Spirtes, **C**lark Glymour): start with all six links; test given **no** friend, then **one**, then **two**.

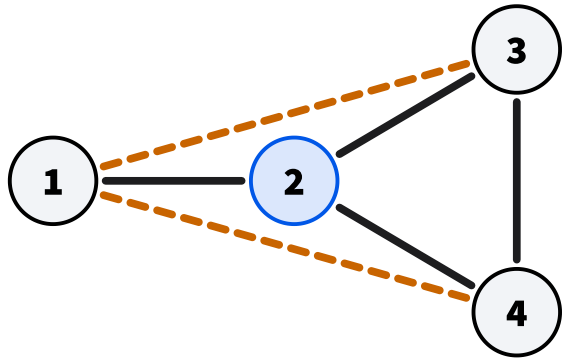
WHY If **any** set of friends explains a tie, the link is not needed; small sets need few days.

round 0: each pair alone



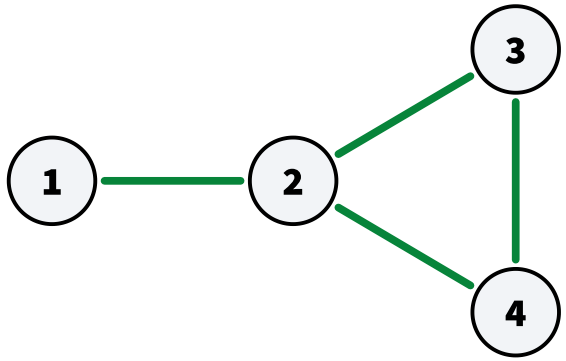
all six tied: keep all

round 1: given one friend



friend 2 explains 1–3 and 1–4

round 2: given two friends



nothing more explained

Guarantee (Spirtes, Glymour; Kalisch, Bühlmann). If every independence of p shows in its map (**faithfulness**) and every test is right, the PC algorithm returns the true links; as N grows, the tests are right with chance tending to 1.

SO **The four true links remain, loop included.**

NEXT Or judge whole maps at once, as with trees?

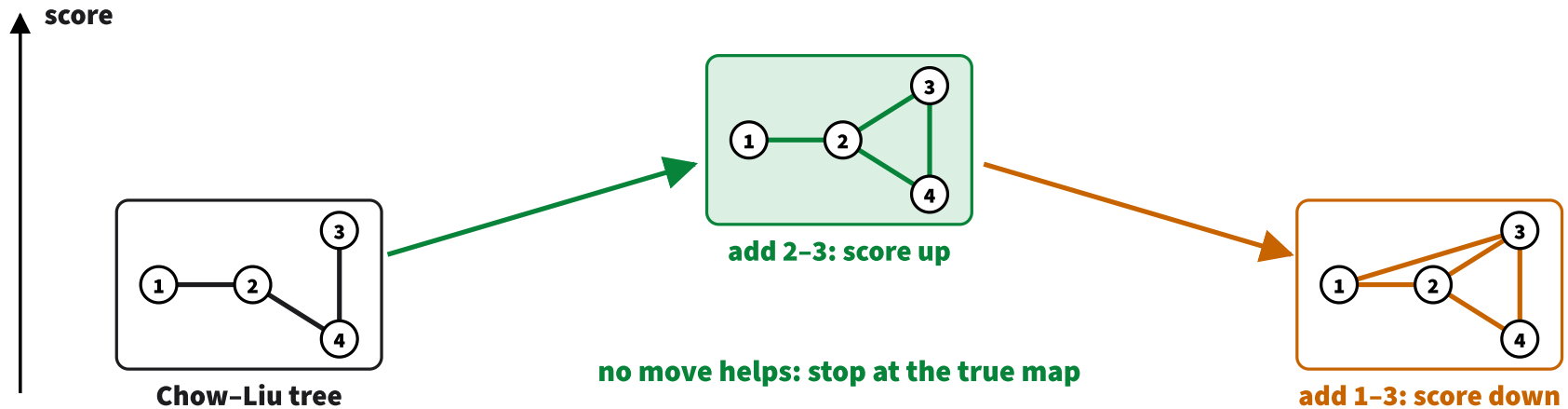
Or score every map, and search by small moves

PROBLEM PC decides **one link at a time**; one wrong test spoils the map.

WE DO Give each **whole map** one score, then climb.

WHY Fit alone always prefers more links (Lecture 7, p. 38); the **penalty** stops that.

1. **Score** = fit to the notebook − a cost for each extra number (Lecture 7, p. 37).
2. **Start** from the Chow–Liu tree.
3. **Try** every one-link change; take the best if the score rises; **stop** when none does.



SO **On the loop, one move adds the missing link 2–3. Cost: with a loop, scoring needs inference (Lecture 6), not counting.**

NEXT Does the climb find the best map?

Theorem F: finding the best map with loops is hard

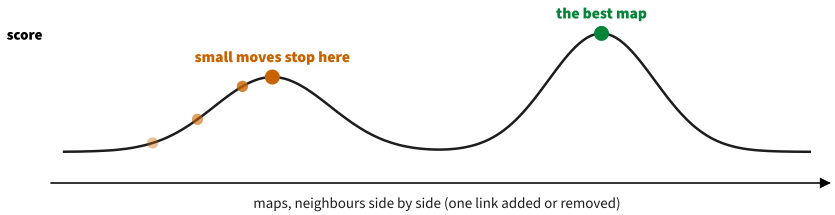
PROBLEM One-link moves can stop where no single move helps.

WE DO Ask whether any fast method finds the best map.

WHY If none does, we must accept good maps, not best ones.

Theorem F (Chickering). Finding the highest-scoring map with arrows is **NP-hard**, even with at most two parents per friend.

In words: beyond trees, the climb may stop only locally best.



Guarantee (Chickering, 2002). Climb in two phases: first **only add** links while the score rises, then **only remove** links while it rises. With **enough days** and a **faithful** p **(Lecture 7, p. 52)**, the climb **ends at the true map** (arrows only as far as data can show them **(Lecture 7, p. 41)**).

SO **Worst case: no fast method is known to find the best score. Yet with enough days, the two-phase climb ends at the truth. Trees: greedy is always exact (Theorem C)** **(Lecture 7, p. 29)****.**

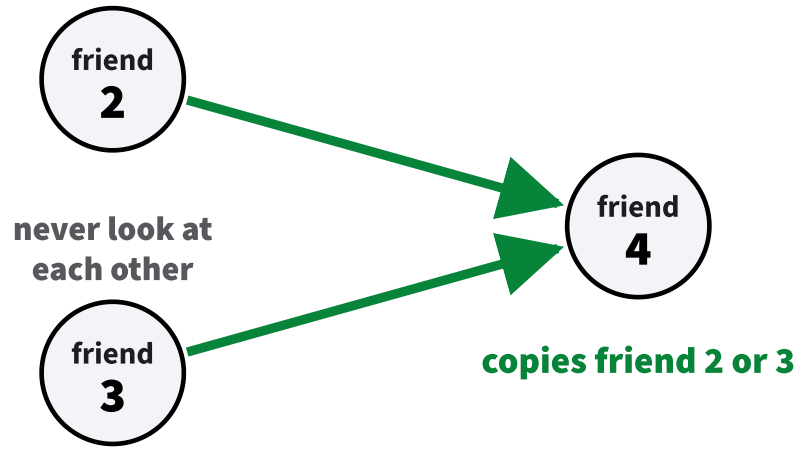
NEXT Can loops at least show us arrows?

Two unlinked friends become linked given a third

PROBLEM Links alone never show arrows (Lecture 7, p. 41).

WE DO Let friends 2 and 3 toss **alone**; friend 4 copies one of them.

WHY Two causes meeting in one friend behave unlike a chain.



friend 4 wrote H, and friend 2 wrote T:

friend 4 probably copied friend 3,
so friend 3 probably wrote H

known friend 4: 2 and 3 become linked

SO **Unlinked alone, linked given friend 4: explaining away.**

NEXT Does this pattern reveal the arrows?

A collider shows the arrows: both point into friend 4

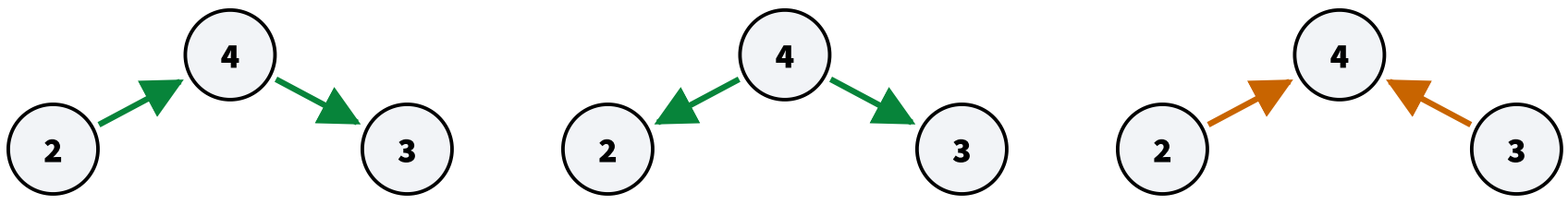
PROBLEM Which arrows can any notebook see?

WE DO Compare chain, fork and collider on three friends.

WHY Only shapes that give different chances can be told apart.

Theorem G (Verma and Pearl). Two arrow maps give the same models exactly when they share links and **colliders** (two arrows into one friend from unlinked friends).

In words: colliders are the arrows the data can see.



same links, same chances: the arrows are hidden

2 and 3 unlinked alone: visible

SO **A chain or a fork hides its arrows; a collider shows them.**

NEXT Collect Part 1, then run every step on the friends' notebook.

Measure, score, build, check, go beyond trees

step	tool	key fact
1 measure	mutual information $\hat{I}$	zero means no link
2 score	Theorem B	best tree = most information
3 build	greedy rule (Kruskal)	exact for trees: Theorem C
4 check	penalty, resampling, new days	short notebooks invent links
5 loops	conditional information, tests, search	test a link given the others

Next: every step on the friends' notebook, with numbers.

Every step of Part 1 runs on the friends' notebook

1. the friends' **true habits**, and a **perfect notebook** of 512 days;
2. each pair's table and information, then the **greedy rule**, step by step;
3. the fitted tables and the scores; then a **short notebook** of 20 days.

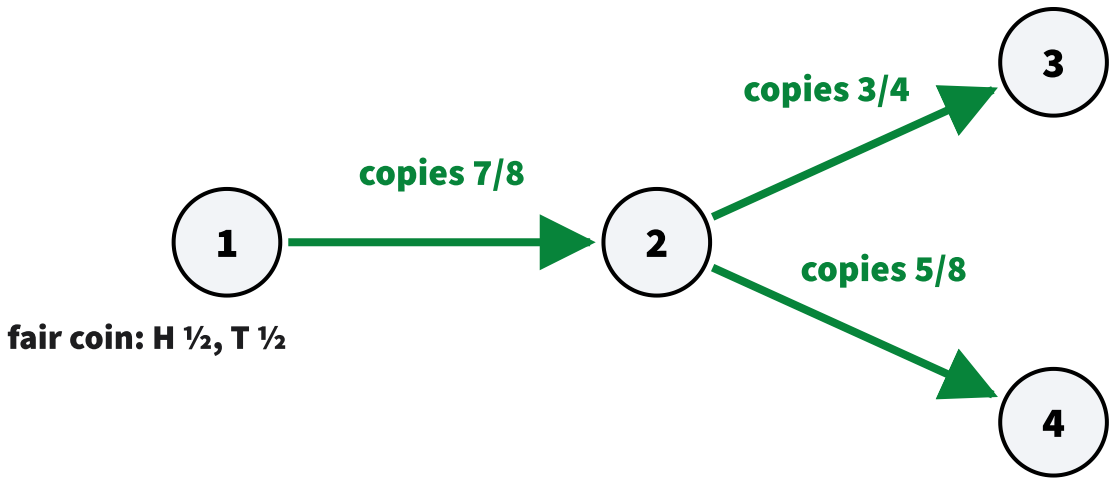


count, measure, sort, link: no search over trees

Every step of Part 1, with every number shown.

The friends' true habits, which the method never sees

friend	habit	why this number
1	tosses a fair coin	no face is preferred
2	writes friend 1's face with chance $\frac{7}{8}$, else the other	a strong link
3	writes friend 2's face with chance $\frac{3}{4}$, else the other	medium: 1–3 stays fairly strong
4	writes friend 2's face with chance $\frac{5}{8}$, else the other	weak: below the indirect pair 1–3

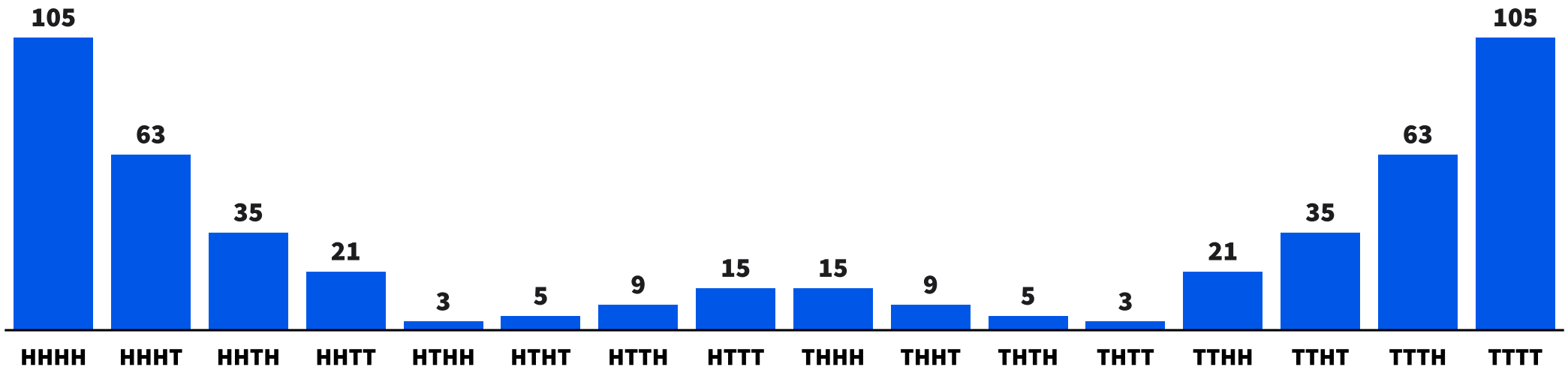


We use the habits only to build the notebook and to grade the answer.

A perfect notebook of 512 days

No chance errors: each kind of day appears **exactly 512 × its chance** times, as in Lecture 5. For example:

$$n(\text{HHHH}) = 512 \cdot \underbrace{\tfrac{1}{2}}_{\text{1 tosses H}} \cdot \underbrace{\tfrac{7}{8}}_{\text{2 copies 1}} \cdot \underbrace{\tfrac{3}{4}}_{\text{3 copies 2}} \cdot \underbrace{\tfrac{5}{8}}_{\text{4 copies 2}} = 105$$



the faces of friends 1, 2, 3, 4 on a day, and how many of the 512 days look like that

Here the counted fractions equal the true chances: no chance errors yet.

EXAMPLE · ONE PAIR'S TABLE

Friends 1 and 2 show the same face on 448 days

Sort the 512 days by the faces of **friends 1 and 2 only**, adding over friends 3 and 4:

friends 1 and 2: counts in 512 days

	friend 2: H	friend 2: T
friend 1: H	224	32
friend 1: T	32	224

same face: 224 + 224 = 448 days

the cell H, H adds the 4 days with friends 1, 2 both H:

HHHH: 105
HHHT: 63
HHTH: 35
HHTT: 21

sum: 224

Each friend shows H on $224 + 32 = 256$ days, half of them: $\hat{p}(x_1{=}\mathrm{H}) = \hat{p}(x_2{=}\mathrm{H}) = \frac{1}{2}$.

One pass over the notebook gives one pair's table.

EXAMPLE · ONE PAIR'S INFORMATION

Friends 1 and 2 share 0.316 nats of information

One term per cell: the counted fraction times the log of (counted fraction ÷ the fraction with no link, $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$) **(Lecture 7, p. 12)**.

faces of 1, 2	counted fraction	with no link	term
H, H	$224/512 = \frac{7}{16}$	$\frac{1}{4}$	$\frac{7}{16} \cdot \ln \frac{7/16}{1/4} = \frac{7}{16} \cdot \ln \frac{7}{4} = 0.245$
T, T	$\frac{7}{16}$	$\frac{1}{4}$	0.245
H, T	$32/512 = \frac{1}{16}$	$\frac{1}{4}$	$\frac{1}{16} \cdot \ln \frac{1/16}{1/4} = \frac{1}{16} \cdot \ln \frac{1}{4} = -0.087$
T, H	$\frac{1}{16}$	$\frac{1}{4}$	-0.087

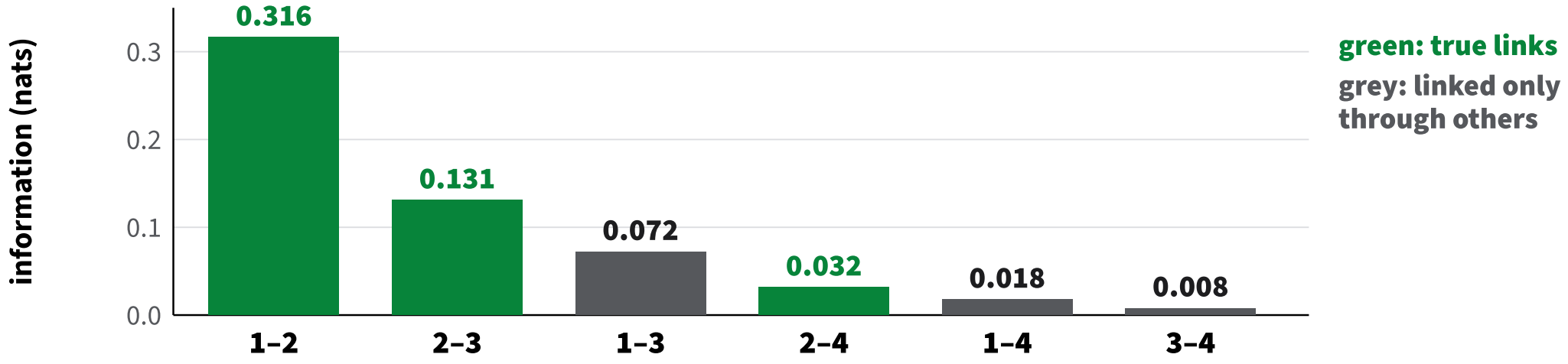
$$\hat{I}(X_1; X_2) = 0.245 + 0.245 - 0.087 - 0.087 = \mathbf{0.316} \text{ nats}$$

Agreeing cells add information; disagreeing cells take a little back.

EXAMPLE · ALL SIX PAIRS

All six pairs: one indirect pair beats a true link

Each friend shows H half the time, so the same-face count gives the information (Lecture 7, p. 62). Same face: 1–2 on 448, 2–3 on 384, 1–3 on 352, 2–4 on 320, 1–4 on 304, 3–4 on 288 days.



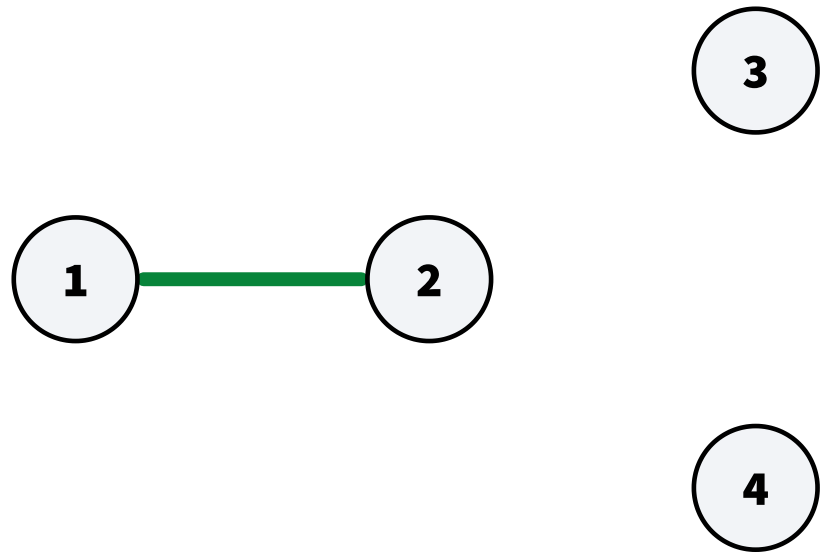
Theorem A holds: 1–3 (0.072) is below 1–2 and 2–3. But it is **above the true link 2–4** (0.032).

Keeping the top three pairs would choose 1–3 and miss 2–4.

Step 1: friends 1 and 2 share the most, so we link them

The strongest pair, 1–2 (0.316), cannot close a loop: there are no links yet.

	pairs, strongest first	
1–2	0.316	✓ link
2–3	0.131	
1–3	0.072	
2–4	0.032	
1–4	0.018	
3–4	0.008	

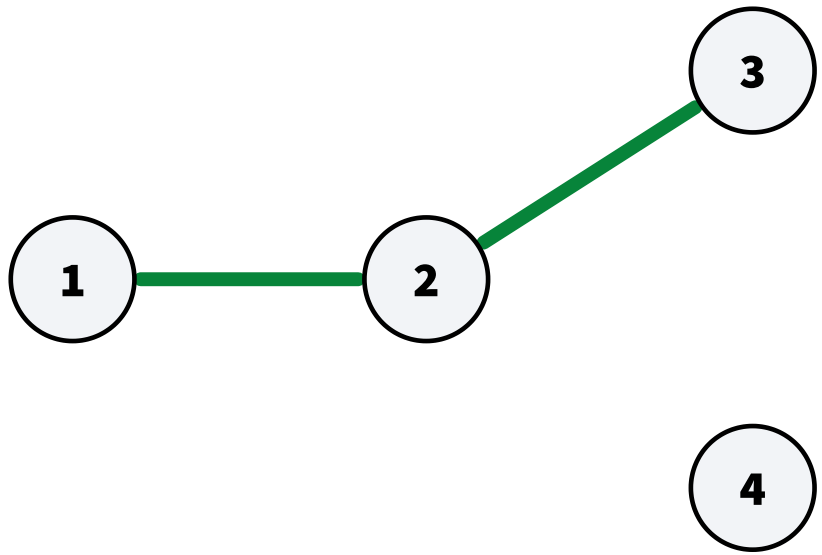


Link 1–2.

Step 2: friends 2 and 3 come next, and no loop forms

Pair 2–3 (0.131) joins friend 3 to the group of friends 1 and 2.

	pairs, strongest first	
1–2	0.316	✓ link
2–3	0.131	✓ link
1–3	0.072	
2–4	0.032	
1–4	0.018	
3–4	0.008	

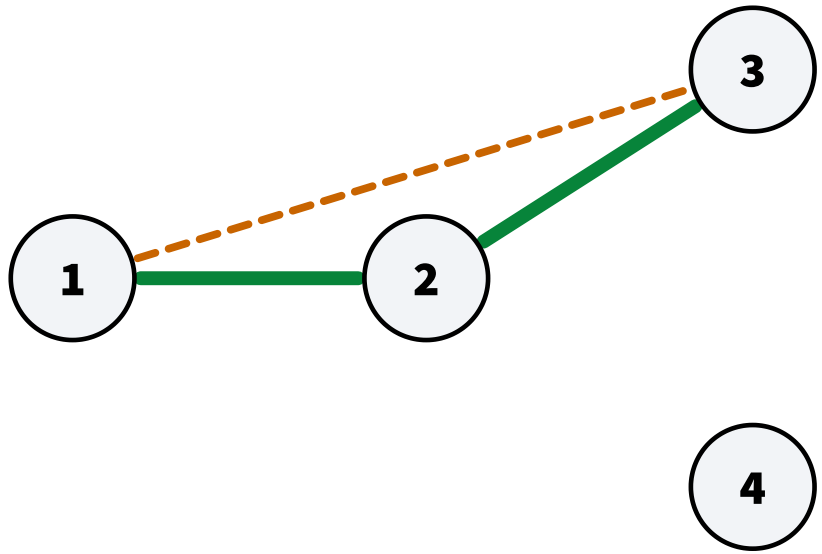


Link 2–3.

Step 3: friends 1 and 3 would close a loop, so we skip

Pair 1–3 (0.072) is strongest left, but 1 and 3 are **already connected** through 2. It would close loop 1–2–3, where it is weakest (Theorem A).

	pairs, strongest first	
1–2	0.316	✓ link
2–3	0.131	✓ link
1–3	0.072	✗ closes a loop
2–4	0.032	
1–4	0.018	
3–4	0.008	

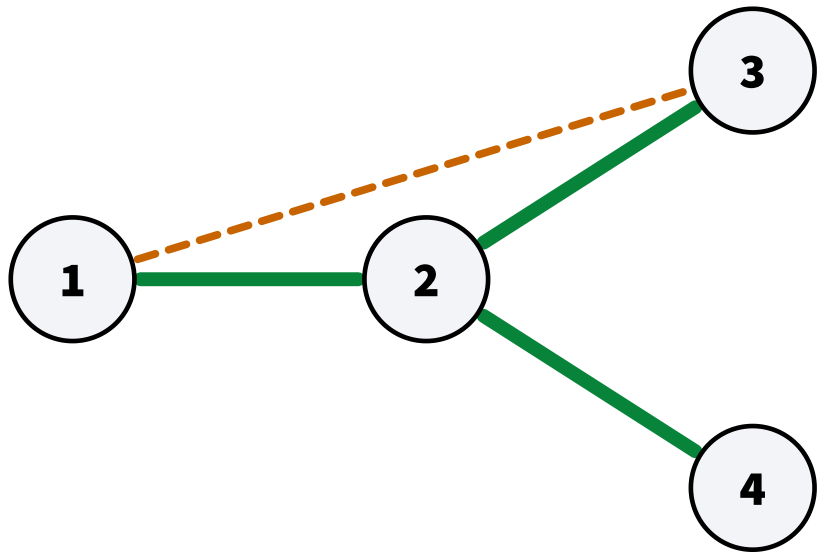


Skip 1–3: the loop check catches the indirect pair.

Step 4: pair 2–4 joins friend 4; three links, done

Pair 2–4 (0.032) joins friend 4. With $3 = 4 - 1$ links everyone is connected, so pairs 1–4 and 3–4 are not needed.

	pairs, strongest first	
1–2	0.316	✓ link
2–3	0.131	✓ link
1–3	0.072	✗ closes a loop
2–4	0.032	✓ link
1–4	0.018	not needed: three links made
3–4	0.008	not needed: three links made



The learned tree is the true tree: 1–2, 2–3, 2–4.

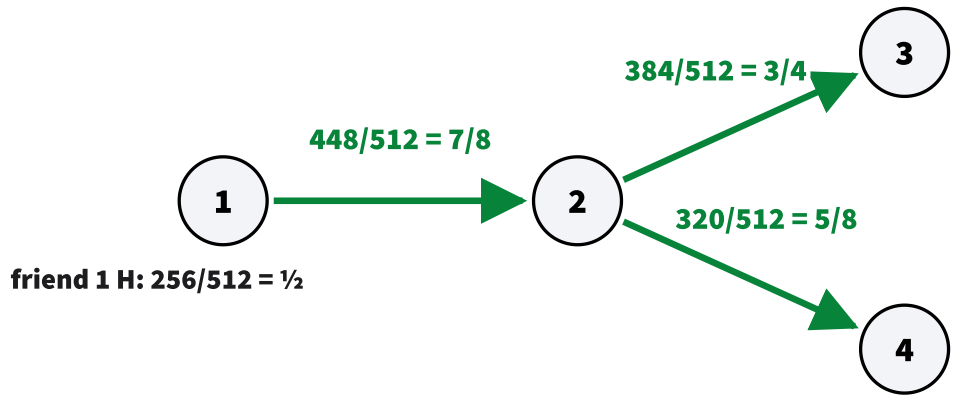
Counting along the learned links gives back the habits

PROBLEM We have the **links**; we still need each friend's **copy chance**.

WE DO For each link, count the days the two friends wrote the **same face**.

WHY The fitted chance is the **share of days** (Lecture 7, p. 19).

link	same face on	share
1–2	448 of 512 days	7/8
2–3	384 of 512 days	3/4
2–4	320 of 512 days	5/8

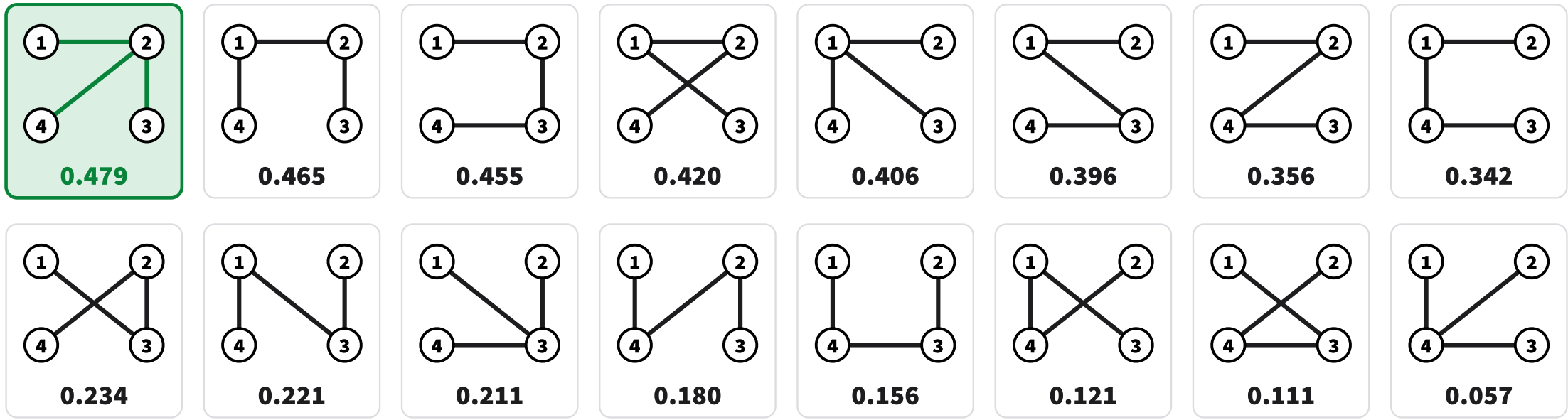


SO **Exactly the friends' habits: a perfect notebook gives back the links and the habits.**

NEXT Is the true tree also the best-scoring of all 16 trees?

The true tree scores best of all 16 trees

Each tree's **total information**, best first. Theorem B: score = total $-\,4\ln 2$, since each friend's face is fair: $\hat{H}(X_i) = \ln 2$.



Best: $0.479 - 2.773 = \mathbf{-2.294}$ per day. With no links: -2.773.

The greedy tree is the best of the 16, as Theorem C promised.

EXAMPLE · FIT AGAINST NUMBERS

The full table ties the tree, but needs 15 numbers, not 7

The notebook comes from a tree, so the full table fits **no better**. The penalty charges $\ln 512/1024 = 0.0061$ per number, per day **(Lecture 7, p. 37)**:

map	numbers	score per day	after the penalty
no links	**4**	**−2.773**	**−2.797**
the true tree	**7**	**−2.294**	**−2.336**
the full table	**15**	**−2.294**	**−2.385**

Same fit, fewer numbers: with the penalty, the tree wins.

Now a real, short notebook: 20 days at random

Draw 20 days from the true habits (one random draw). Now the counted fractions **miss** the true chances.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
friend 1	H	H	H	H	H	H	T	H	H	H	T	H	T	T	T	T	H	T	H	T
friend 2	H	H	H	H	T	H	T	H	T	H	T	T	T	T	T	T	H	H	H	T
friend 3	H	H	H	H	T	T	T	H	T	H	T	T	T	H	T	T	H	H	H	H
friend 4	H	T	H	H	T	T	T	T	T	H	T	H	T	H	H	H	H	H	T	T

20 days, drawn at random with the friends' true habits (one draw)

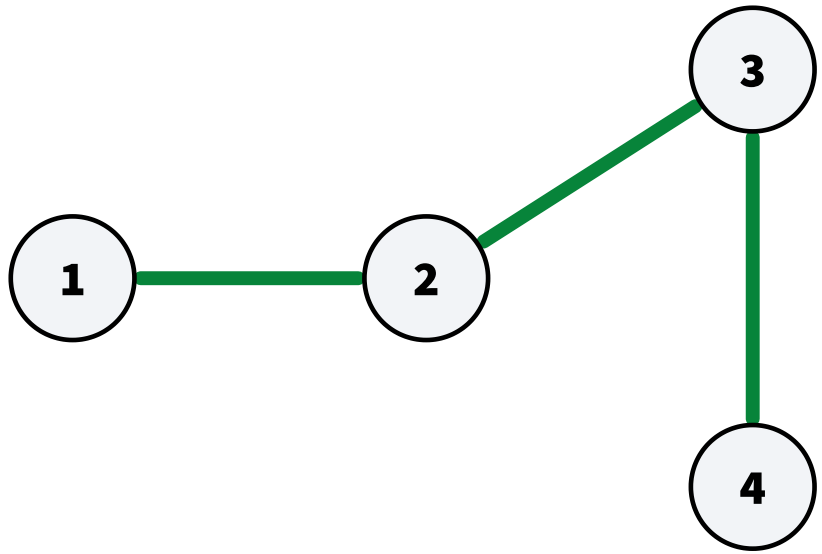
The same habits, but only 20 days of evidence.

On 20 days, greedy links friend 4 to friend 3

By chance, pair 3–4 (0.046) comes out above the true link 2–4 (0.020), so the greedy rule takes it.

pairs in the 20-day notebook, strongest first

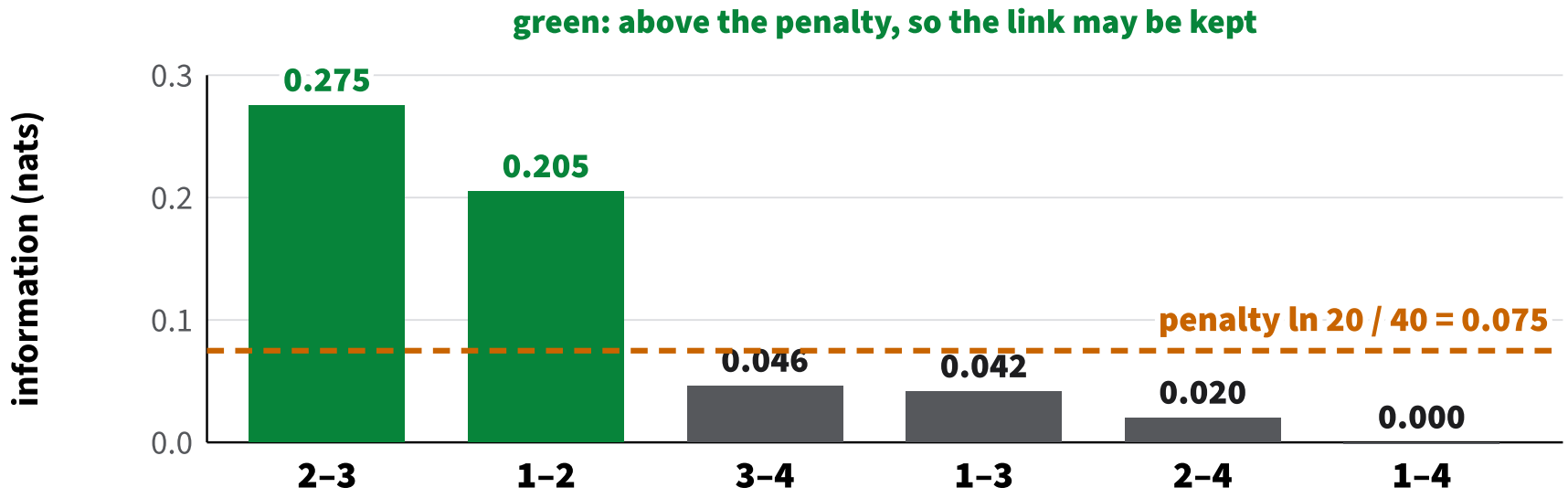
2–3	**0.275**	**✓ link**
1–2	**0.205**	**✓ link**
3–4	**0.046**	**✓ link**
1–3	**0.042**	not needed: three links made
2–4	**0.020**	not needed: three links made
1–4	**0.000**	not needed: three links made



Right links for friends 1, 2, 3; the wrong one for friend 4.

The penalty leaves friend 4 unlinked after 20 days

The threshold is $\ln 20/40 = 0.075$ **(Lecture 7, p. 37)**. Links 2–3 and 1–2 clear it; 3–4 (0.046) does not.



Friend 4 stays **unlinked**: "not enough evidence", not a wrong link.

The penalty turns a wrong link into a visible gap.

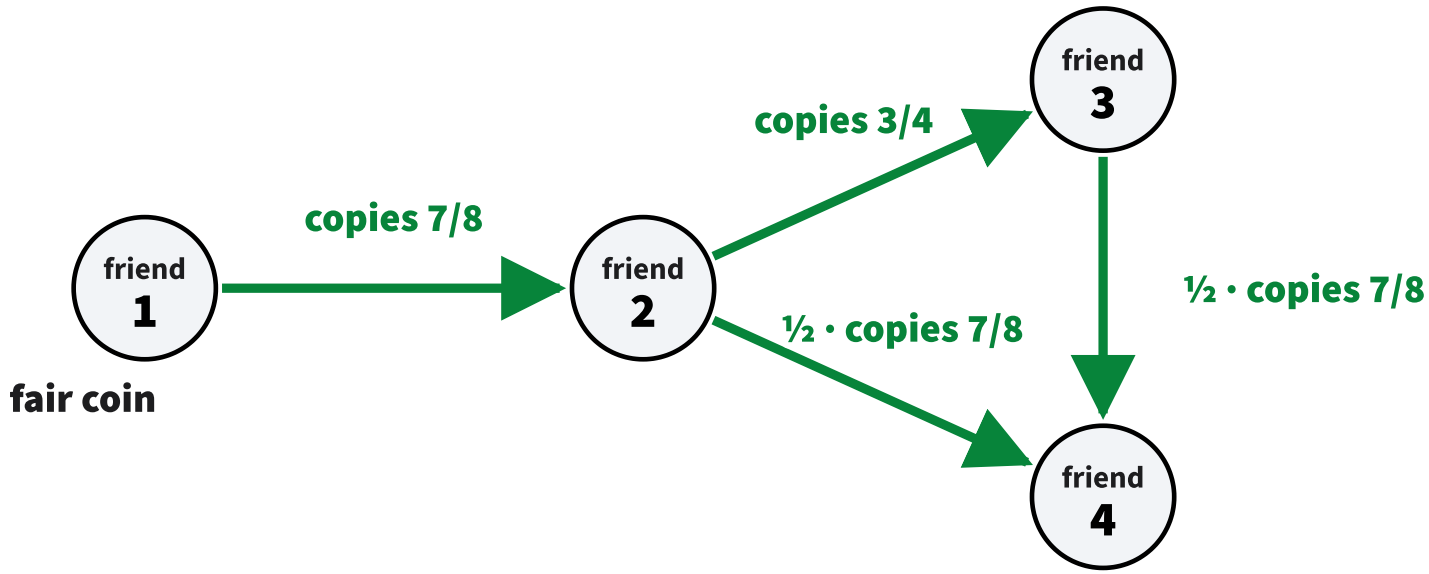
Longer notebooks place friend 4 correctly

notebook	links found	friend 4
512, perfect	1–2, 2–3, 2–4: all true	right, habits exact
40, one draw	1–2, 2–3, 3–4	wrong; resampling flags it (Lecture 7, p. 39)
20, one draw	2–3, 1–2, 3–4	wrong; the penalty drops it
320, 1000 draws	true tree in 93%	most errors misplace friend 4 (Lecture 7, p. 36)

Long notebooks give the truth; short ones need the checks.

A loop: friend 4 copies friend 2 or friend 3

Friends 1, 2, 3 as before. A fair toss tells friend 4 whom to copy; it copies with chance 7/8. A **perfect notebook** of 512 days **(Lecture 7, p. 107)**.

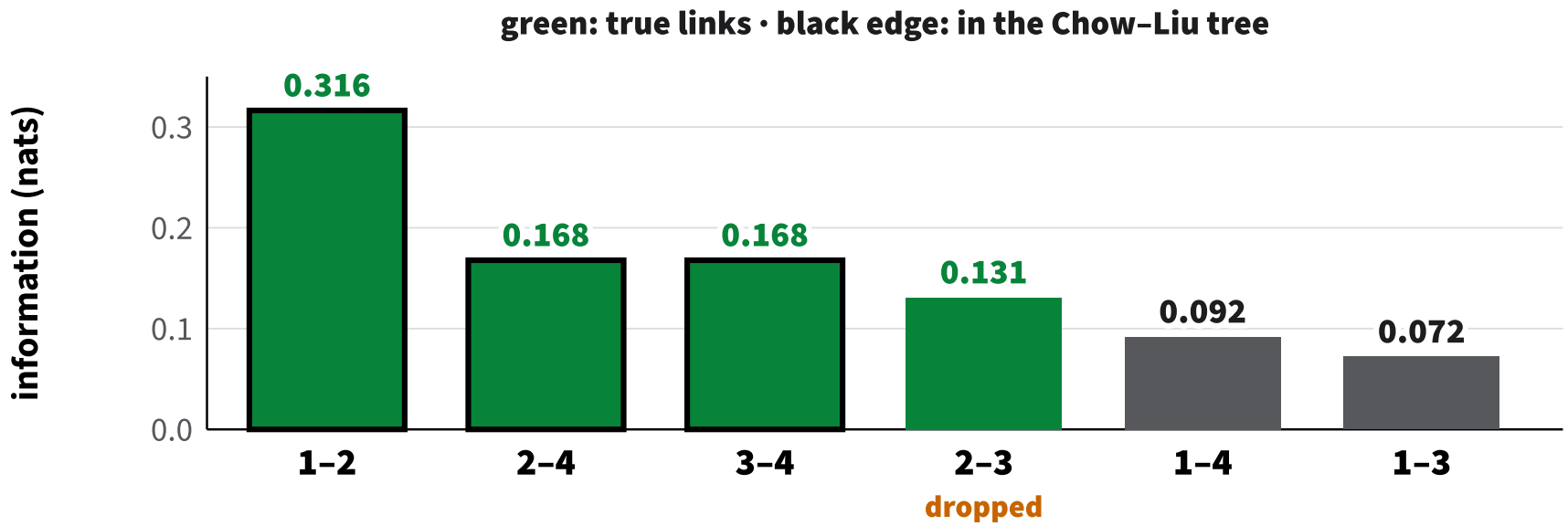


Four true links: 1–2, 2–3, 2–4, 3–4.

EXAMPLE · A LOOP

The Chow–Liu tree drops the true link 2–3

Pairs 2–4 and 3–4 tie at 0.168 and enter after 1–2. Three links are done, so 2–3 (0.131) is never taken.



A tree keeps three links; this loop needs four.

EXAMPLE · BY HAND

Given friends 2 and 3, friend 1 says nothing about friend 4

Take one group of days and compare the two rows of each table **(Lecture 7, p. 50)**:

friends 1 and 4

in the group: 2 = H, 3 = H

	friend 4: H	friend 4: T
friend 1: H	147	21
friend 1: T	21	3

rows in the same ratio, 7 : 1: no link here

friends 3 and 4

in the group: 1 = H, 2 = H

	friend 4: H	friend 4: T
friend 3: H	147	21
friend 3: T	28	28

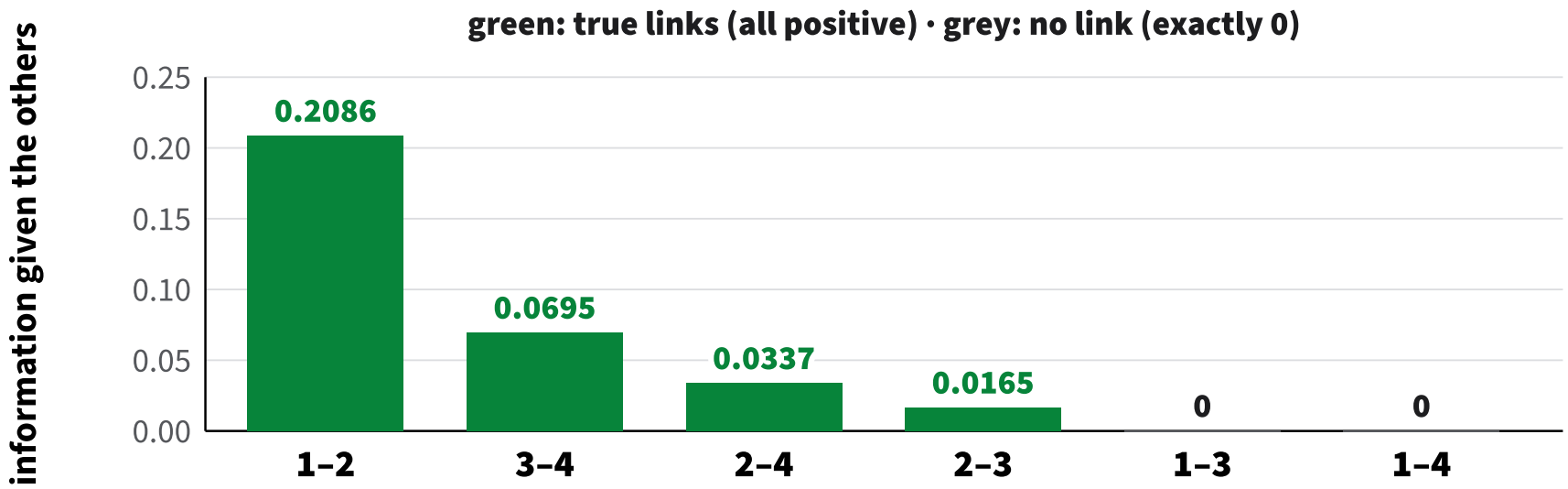
rows 7 : 1 and 1 : 1: linked

Same ratio in both rows: no information. Different ratios: a link.

EXAMPLE · A LOOP

Given the others, exactly the four true links remain

Every pair, all four groups: conditional information is **0 for 1–3 and 1–4**, positive for the four true links.



Link 2–3 is faintest, 0.0165: it needs the most days **(Lecture 7, p. 51)**.

Conditional information finds the loop that the tree missed.

A collider: 2 and 3 are unlinked until friend 4 is known

PROBLEM Friends 2 and 3 toss **alone**; friend 4 copies one of them (chance 7/8).

WE DO Guess friend 3's face from friend 4, then **learn friend 2** too.

WHY If learning 2 changes our guess about 3, they are **linked given 4**.

all 256 days

	friend 3: H	friend 3: T
friend 2: H	64	64
friend 2: T	64	64

every cell 64: no link, information 0

the 128 days where friend 4 wrote H

	friend 3: H	friend 3: T
friend 2: H	56	32
friend 2: T	32	8

rows 1.75 : 1 and 4 : 1: linked

1. **Friend 4 wrote H.** Friend 3 wrote H on 88 of those 128 days: **0.69**.
2. **Also friend 2 wrote T.** Then 4's H likely came from **3**: friend 3 H on 32 of 40 days, **0.8**.

SO **Friend 2 explains away part of 4's H: given 4, 2 and 3 are linked (conditional information 0.014)** **(Lecture 7, p. 108).**

NEXT The same question inside today's artificial intelligence (AI).

Next: the same question inside today's AI

Friends become **words, records, balls or variables**; the question stays: who is linked to whom?



from the notebook alone, to a map of links we can trust

A language model links each word to other words. How often does it choose these links?

once, after training | anew for every sentence | never

Answer. Anew for every sentence: attention, the last slide of Part 3.

Learning links is everywhere in machine learning.

Learning links runs inside many systems

Classifiers	Mixtures of trees	Physics from motion	Learning arrows	Attention
link the features with a tree	each record from one of several trees (Lecture 5's EM	a neural network guesses the springs	"no loop of arrows" as one equation	new links for every sentence

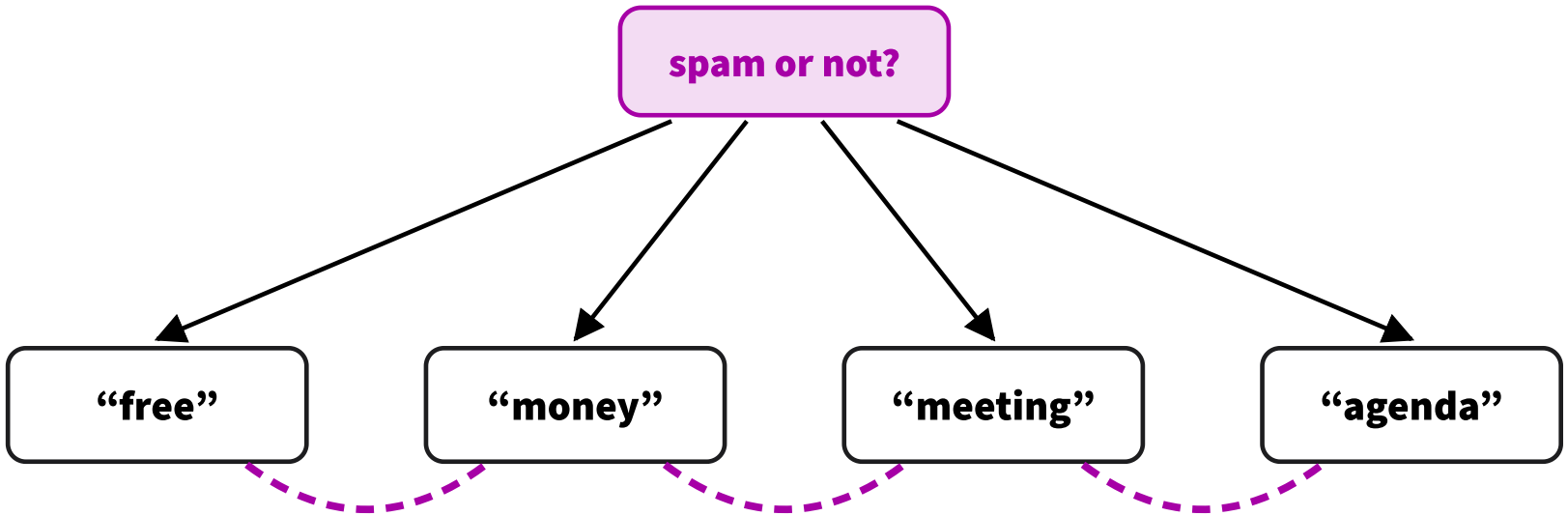
the same question, who is linked to whom, at scale

Only the friends change: words, records, particles, variables.

EM = expectation–maximization · **AI** = artificial intelligence

A spam filter can link words that appear together

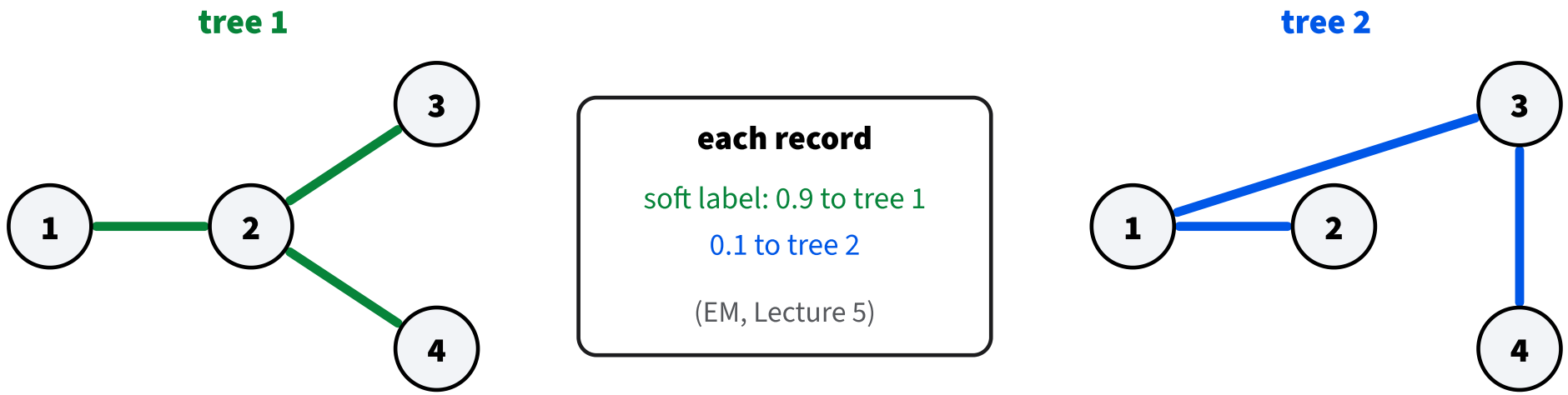
1. Friends = **words** of an email; notebook = emails marked spam or not.
2. **Naive Bayes**: words unlinked given the class, so "free" and "money" count **twice**.
3. **Tree-augmented naive Bayes** adds a Chow–Liu tree on the words, using information **given the class**.



One extra tree of links, and evidence is no longer counted twice.

When one tree is not enough, EM mixes several trees

1. Records follow different maps: a **hidden choice of tree**, like Lecture 5's hidden coin; fit it with Lecture 5's **expectation–maximization** (EM).
2. **Guess step:** soft labels say which tree made each record.
3. **Count step:** Chow–Liu per tree, weighted by the soft labels.

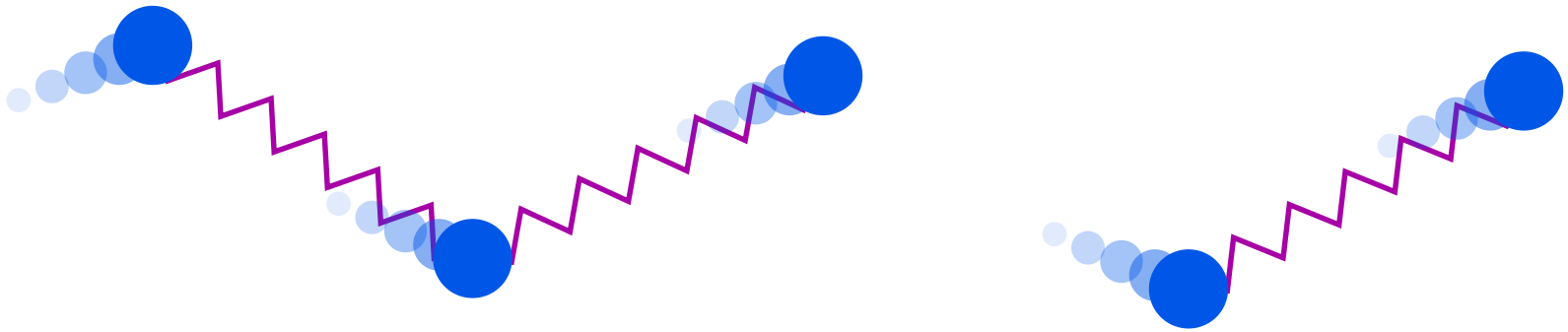


then Chow–Liu on each tree's weighted notebook, and repeat

Lecture 5's EM plus today's Chow–Liu: a mixture of trees.

A neural network learns which balls are tied by springs

1. Friends = **moving balls**; notebook = their trails.
2. Links = **hidden springs**. One network guesses them; a second predicts the motion.
3. Both train on Lecture 6's **evidence lower bound**, with links as hidden variables.



we see the trails; the springs (magenta) are hidden links a neural network learns to guess

Neural relational inference learns the links with Lecture 6's score.

One equation forbids loops, so gradients can learn arrows

PROBLEM Searching arrow maps one move at a time is **slow** and can stop early (Theorem F).

WE DO Write the map as weights W ($W_{ij} \neq 0$: arrow $i \to j$); turn "**no loop of arrows**" into one equation.

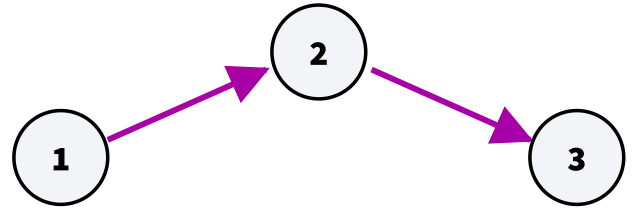
WHY Then **gradient descent**, as for neural networks, searches all maps at once.

$$\text{no loop of arrows} \iff h(W) = \operatorname{tr}\left(e^{W \circ W}\right) - m = 0$$

$W \circ W$: squared entries; $(W \circ W)^k$ has a positive diagonal entry exactly when a loop of length k exists; $e^A = I + A + A^2/2! + \cdots$ adds all lengths; tr sums the diagonal. No loop: diagonal of e^A is all 1, trace $= m$ friends.

weights W: row acts on column

	1	2	3
1	0	0.8	0
2	0	0	0.5
3	0	0	0



no directed loop ⇔ tr(exp(W∘W)) − m = 0

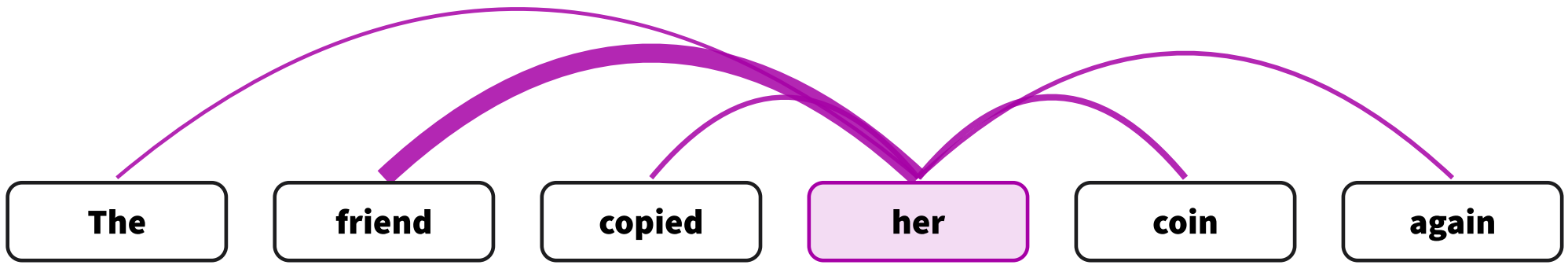
one smooth equation: gradient methods can search

SO **NOTEARS: maximize fit − penalty subject to $h(W) = 0$. A good map, not always the best; arrows still need assumptions (e.g. equal noise).**

NEXT Can links change with every input?

Attention chooses new links for every sentence

1. Friends = **words** of one sentence.
2. Each word weights every (earlier) word: a soft map, **computed anew for each sentence**.
3. Today: one map per notebook. Attention learns **how to draw a map** per input.



“her” links most strongly to “friend”: a new map of links for each sentence (schematic)

From one learned map to a new map for every input.

From a notebook to a map of links we can trust

Who copies whom? We see only the notebook; maps are too many to try.

1. **Measure** each pair: mutual information, zero for no link.
2. **Score** a tree: the information on its links (Theorem B).
3. **Build** the best tree: strongest first, never a loop (Theorem C).
4. **Check** it: penalty, resampling, new days.
5. **Beyond trees:** test a link given the others; colliders show arrows.

On a tree, data give **links, not arrows**; a hidden friend **fakes links**.

Measure, score, build, check.

Check yourself (1 of 3): measuring and building

1. Two friends' counted information over 30 days is 0.02. Are they linked?

yes, surely | not necessarily | no, never

Answer. Not necessarily: chance alone gives about $1/(2N) \approx 0.017$.

2. Pair 1–3 has more information than pair 2–4, yet the greedy rule skips 1–3. Why?

it closes a loop of stronger links | it is too weak | it is chosen at random

Answer. It closes a loop 1–2–3, whose other links are stronger.

Chance makes links look real; loops reveal indirect pairs.

Check yourself (2 of 3): scores and arrows

3. We join two separate pieces of a map with one more link. The notebook's score …

falls | can go either way | never falls

Answer. **Never falls:** it rises by the link's information, so we need a penalty.

4. Can the notebook tell whether friend 2 copies friend 1, or friend 1 copies friend 2?

yes, from the counts | no, both give the same chances | only with 512 days

Answer. **No:** both starting friends give every day the same chance.

More links always fit better; on a tree, arrows need an experiment.

Check yourself (3 of 3): beyond trees

5. Friends 1 and 4 have information 0.09 between them, yet need no link. How can that be?

the notebook is wrong | given friends 2 and 3, it drops to 0 | 0.09 is too small to count

Answer. Given friends 2 and 3 it is 0: they carry all the news between 1 and 4.

6. Friends 2 and 3 are independent, but dependent once friend 4 is known. Which arrows does this show?

2 → 3 | 2 → 4 → 3 | 2 → 4 ← 3

Answer. 2 → 4 ← 3: a collider, the arrows the data can see.

Test links given the others; colliders show arrows.

References: trees, spanning trees and information

[Theorem B, Chow–Liu trees] C. K. Chow and C. N. Liu. Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory* 14(3):462–467, 1968.

[Theorem D, consistency] C. K. Chow and T. J. Wagner. Consistency of an estimate of tree-dependent probability distributions. *IEEE Transactions on Information Theory* 19(3):369–371, 1973. · V. Y. F. Tan, A. Anandkumar, L. Tong and A. S. Willsky. A large-deviation analysis of the maximum-likelihood learning of Markov tree structures. *IEEE Transactions on Information Theory* 57(3):1714–1735, 2011.

[Theorem C, Kruskal] J. B. Kruskal. On the shortest spanning subtree of a graph and the traveling salesman problem. *Proceedings of the American Mathematical Society* 7(1):48–50, 1956.

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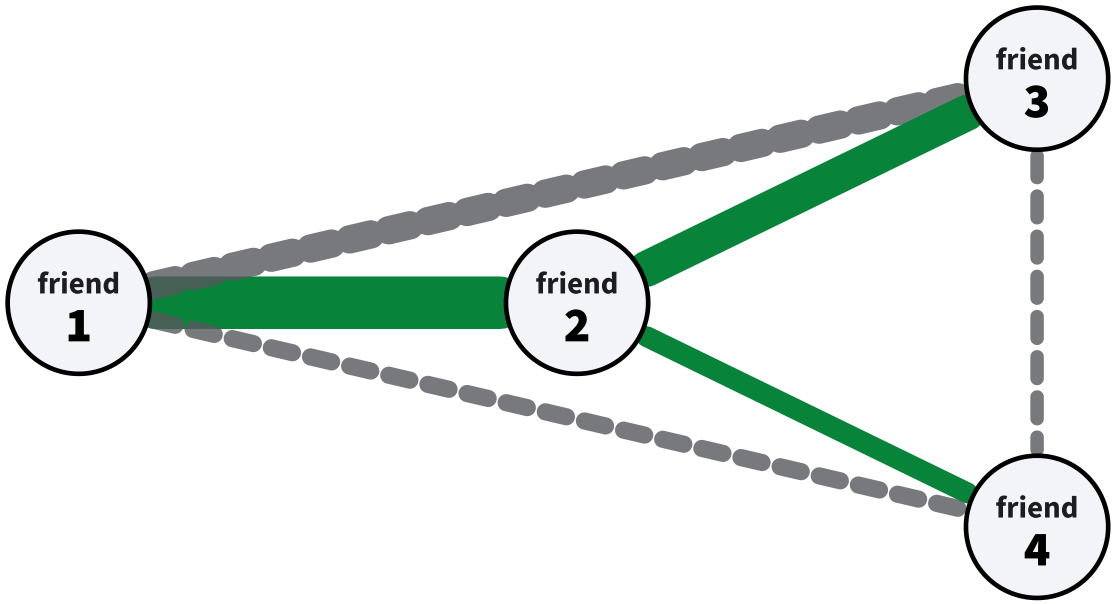
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NEXT WEEK

Next: the midterm, then learning the message rule

Thursday 29 October, 9:30–11:00, W1-233: midterm, open book, **Lectures 1 to 6**. Today is **not examined**.



green: true links · grey dashed: pairs linked only through others · thickness: strength

Week 9, neural belief propagation: keep the map, and **learn the message rule** itself.

> **Today we learned the links. Next we learn the messages.**

The appendix: every proof the lecture pointed to

Measure and score

Theorem A's proof; counted tables are best; the score as a distance; every starting friend gives the same model.

Trust

Why fake links average $1/(2N)$; the penalty per link; the hidden friend's numbers.

The perfect notebook

All 16 counts, and how each is made.

The random notebooks

The 20-day tables; recovery, training and new-day scores for every length.

Nothing here is needed to follow the lecture; everything here supports a claim made in it.

Proof of Theorem A: count the same information two ways

↳ **Chain rule** (Lecture 3): what two faces tell together = what the first tells + what the second adds given the first. **Markov chain:** once x_2 is known, x_3 tells nothing more about x_1.

$$I(X_1; X_2, X_3) = I(X_1; X_2) + I(X_1; X_3 \mid X_2) = I(X_1; X_2)$$

chain rule; the last term is 0 on a chain

$$I(X_1; X_2, X_3) = I(X_1; X_3) + I(X_1; X_2 \mid X_3)$$

chain rule, in the other order

$$I(X_1; X_3) = I(X_1; X_2) - I(X_1; X_2 \mid X_3) \;\leq\; I(X_1; X_2)$$

subtract; information is never negative

The chain read backwards, $X_3 \to X_2 \to X_1$, is also a Markov chain, so the same steps give $I(X_1; X_3) \leq I(X_2; X_3)$.
Our friends: 1 and 3 show the same face when both copies are right or both wrong: $\frac{7}{8} \cdot \frac{3}{4} + \frac{1}{8} \cdot \frac{1}{4} = \frac{11}{16}$, information 0.072, below 0.131 and 0.316.

Given the middle friend, the far friend adds nothing, so information cannot grow.

Proof: on a fixed tree, counted tables give the best fit

↳ The tree started at friend 1, with free tables p: $L(p)$ = its log-likelihood of the notebook, so $\ell(\mathcal{T}) = \max_p L(p)/N$ **(Lecture 7, p. 18)**.
Theorem 1 of Lecture 5 **(Lecture 5, p. 11)**: $h \cdot \ln\theta + t \cdot \ln(1-\theta)$ is largest at $\theta = h/(h+t)$.

$$L(p) = \sum_{d=1}^{N} \left(\ln p\big(x_1^{(d)}\big) + \sum_{(i \to j)} \ln p\big(x_j^{(d)} \mid x_i^{(d)}\big) \right)$$

log of a product = sum of logs

$$= \sum_{x_1} n(x_1) \cdot \ln p(x_1) + \sum_{(i \to j)} \sum_{x_i} \sum_{x_j} n(x_i, x_j) \cdot \ln p(x_j \mid x_i)$$

group days by the faces each term reads

$$\text{row } x_i \text{ of link } i \to j: \quad n(x_i, \mathrm{H}) \cdot \ln\theta + n(x_i, \mathrm{T}) \cdot \ln(1-\theta), \;\; \theta = p(\mathrm{H} \mid x_i)$$

each row is a coin; rows share no number

$$\Rightarrow \; \hat{p}(x_j \mid x_i) = \frac{n(x_i, x_j)}{n(x_i)}, \qquad \hat{p}(x_1) = \frac{n(x_1)}{N}$$

Theorem 1 for each coin

Each row of each table is a separate coin, so counting fits them all.

The score is minus an entropy, minus a distance

↳ $\hat{p}(\boldsymbol{x}) = n(\boldsymbol{x})/N$, the counted fraction of whole days; $\hat{H}(\boldsymbol{X}) = -\sum_{\boldsymbol{x}} \hat{p}(\boldsymbol{x}) \cdot \ln \hat{p}(\boldsymbol{x})$; $\mathrm{KL}(p \| q) = \sum p \cdot \ln(p/q)$.

$$\ell(\mathcal{T}) = \sum_{\boldsymbol{x}} \hat{p}(\boldsymbol{x}) \cdot \ln \hat{p}_{\mathcal{T}}(\boldsymbol{x})$$

group equal days

$$= \sum_{\boldsymbol{x}} \hat{p}(\boldsymbol{x}) \cdot \ln \hat{p}(\boldsymbol{x}) - \sum_{\boldsymbol{x}} \hat{p}(\boldsymbol{x}) \cdot \ln \frac{\hat{p}(\boldsymbol{x})}{\hat{p}_{\mathcal{T}}(\boldsymbol{x})}$$

add and subtract $\ln \hat{p}(\boldsymbol{x})$

$$= -\hat{H}(\boldsymbol{X}) - \mathrm{KL}\left(\hat{p} \,\|\, \hat{p}_{\mathcal{T}}\right)$$

definitions

The full table is $\hat{p}$ itself: distance 0, the highest score of all maps. On the perfect notebook the true tree also has distance 0, which is why it ties the full table **(Lecture 7, p. 70)**.

Highest score = smallest distance from the counted days.

Every starting friend gives the same tree model

↳ Each step uses one identity: $p(a) \cdot p(b \mid a) = p(a, b) = p(b) \cdot p(a \mid b)$.

$$p(x_1) \cdot p(x_2 \mid x_1) \cdot p(x_3 \mid x_2) \cdot p(x_4 \mid x_2)$$

start at friend 1

$$= p(x_2) \cdot p(x_1 \mid x_2) \cdot p(x_3 \mid x_2) \cdot p(x_4 \mid x_2)$$

turn the link 1–2 around

$$= p(x_4) \cdot p(x_2 \mid x_4) \cdot p(x_1 \mid x_2) \cdot p(x_3 \mid x_2)$$

turn the link 2–4 around: start at friend 4

$$= \prod_i p(x_i) \cdot \prod_{(i,j) \in \mathcal{T}} \frac{p(x_i, x_j)}{p(x_i) \cdot p(x_j)}$$

write each $p(b \mid a)$ as $p(a, b)/p(a)$, then cancel

Every start gives the same chances, so no notebook can tell who tossed and who copied.

Fake links: about $1/(2N)$ on average

↳ For two friends with fair coins who never copy, $2N \cdot \hat{I}$ is close to a chi-squared variable with one degree of freedom (the likelihood-ratio test of independence), whose average is 1. So $\hat{I} \approx 1/(2N)$ on average.

days N	average $\hat{I}$, exact	$1/(2N)$	chance $\hat{I} \geq 0.0316$ (friend 4's link)
10	0.065	0.050	47%
20	0.028	0.025	29%
40	0.013	0.0125	13%
80	0.0064	0.00625	2.6%

Exact: every count table (a, b, c, d) with $a + b + c + d = N$ is listed, with its multinomial chance $N!/(a!\, b!\, c!\, d!) \cdot (1/4)^N$.

Short notebooks: fake links as strong as weak real ones.

The penalty per link: keep it if $\hat{I} > \ln N/(2N)$

↳ **Bayesian information criterion:** total log-likelihood $-\frac{k}{2} \cdot \ln N$, with k the numbers fitted. Joining two separate pieces by a link adds one number (its table has 2 free numbers, the friend's own face had 1).

$$\Delta\big(N \cdot \ell\big) = N \cdot \hat{I}(X_i; X_j)$$

Theorem B: the score rises by the link's information

$$\Delta\Big(N \cdot \ell - \tfrac{k}{2} \cdot \ln N\Big) = N \cdot \hat{I}(X_i; X_j) - \tfrac{1}{2} \cdot \ln N$$

one more number: $\Delta k = 1$

$$> 0 \iff \hat{I}(X_i; X_j) > \frac{\ln N}{2N}$$

divide by N

So run the greedy rule on the weights $\hat{I} - \ln N/(2N)$ and stop at the first one not above 0: the best forest. $N = 512$: 0.0061; $N = 20$: 0.075; friend 4's true information (0.0316) clears it from 67 days on.

A link must carry more than $\ln N/(2N)$ nats of information.

The hidden friend: three pairs, a star at friend 1

↳ Friend 2 never writes, so only the pairs among friends 1, 3, 4 can be counted (perfect notebook, 512 days).

pair	same face on	information	greedy rule
1–3	**352 of 512**	**0.072**	**link**
1–4	**304 of 512**	**0.018**	**link**
3–4	**288 of 512**	**0.008**	**closes a loop: skip**

The result links 1–3 and 1–4, but in truth friends 1, 3 and 4 are linked **only through friend 2**. A latent tree model places a hidden friend in the middle from the informations, and expectation–maximization (EM) (Lecture 5) fits its tables.

Strengths multiply: if 2 copies 1 with chance a and 3 copies 2 with chance b, then $P(X_1{=}X_3) = a \cdot b + (1-a) \cdot (1-b)$, so $2 \cdot P(X_1{=}X_3) - 1 = (2a-1) \cdot (2b-1)$, i.e. $\rho_{13} = \rho_{12} \cdot \rho_{23}$. Same as "pass on with chance ρ, else a fair toss": then $P(\text{same}) = \rho + (1-\rho)/2 = (1+\rho)/2$.

What we cannot see, we mistake for direct links.

The perfect notebook: all 16 counts

↳ Count = $512 \cdot \frac{1}{2}\cdot$(2 copies 1: $\frac{7}{8}$ or $\frac{1}{8}$)$\cdot$(3 copies 2: $\frac{3}{4}$ or $\frac{1}{4}$)$\cdot$(4 copies 2: $\frac{5}{8}$ or $\frac{3}{8}$).

day	count	day	count	day	count	day	count
HHHH	105	HTHH	3	THHH	15	TTHH	21
HHHT	63	HTHT	5	THHT	9	TTHT	35
HHTH	35	HTTH	9	THTH	5	TTTH	63
HHTT	21	HTTT	15	THTT	3	TTTT	105

Example: in HTHH, friends 2, 3 and 4 all miss: $n(\mathrm{HTHH}) = 512 \cdot \underbrace{\frac{1}{2}}_{1\text{ tosses H}} \cdot \underbrace{\frac{1}{8}}_{2\text{ misses 1}} \cdot \underbrace{\frac{1}{4}}_{3\text{ misses 2}} \cdot \underbrace{\frac{3}{8}}_{4\text{ misses 2}} = 3.$

Sixteen counts, each a product of four named chances.

The 20-day notebook: six pair tables

↳ Counts of (H,H), (H,T), (T,H), (T,T) for each pair, from the 20 days drawn with the true habits **(Lecture 7, p. 71)**.

pair	H,H	H,T	T,H	T,T	information	greedy
2–3	9	1	2	8	0.275	link
1–2	9	3	1	7	0.205	link
3–4	7	4	3	6	0.046	link (wrong)
1–3	8	4	3	5	0.042	not needed
2–4	6	4	4	6	0.020	not needed (true link)
1–4	6	6	4	4	0.000	not needed

By chance, friend 4 agrees with friend 3 more often than with friend 2.

Scores on the notebook and on new days, by length

↳ Average over 2000 notebooks per length. New days: the exact average log-likelihood under the true chances. Tables for new days use pseudo-count 1 (Laplace's rule, Lecture 5) so that no day gets chance 0. Truth: −2.294.

days	10	20	40	80	160	320	640
full table, own notebook	−1.664	−1.934	−2.101	−2.197	−2.244	−2.270	−2.281
tree, own notebook	−1.803	−2.065	−2.192	−2.248	−2.271	−2.283	−2.287
no links, own notebook	−2.559	−2.669	−2.723	−2.748	−2.760	−2.766	−2.769
full table, new days	−2.573	−2.488	−2.416	−2.364	−2.333	−2.315	−2.305
tree, new days	−2.591	−2.471	−2.392	−2.344	−2.319	−2.306	−2.299
no links, new days	−2.931	−2.861	−2.819	−2.796	−2.785	−2.779	−2.776

On its own notebook the full table always looks best, even better than the truth. On new days the tree wins from 20 days on; with only 10, the heavily smoothed full table edges ahead.

Judge a map on days it has not seen.

Theorem E needs every day to have a positive chance

↳ **Theorem E** combines the pairwise Markov property with the Hammersley–Clifford theorem: for a positive model, the pairs dependent given all others are exactly the links of the smallest fitting map.

Why positivity matters: let three friends always write the **same** face. Given any one of them, the other two are fixed, so every pair is independent given the third, and the test removes **every link**, though the faces are tied together.

In our friends every day has a positive chance (no copy is certain), so the test is safe.

With certain copies, conditional tests can erase real links.

Fake conditional information grows with the groups

↳ In each group, two friends who are independent there give $2N_g \cdot \hat{I}_g$ close to a chi-squared variable with one degree of freedom (Section 4, **(Lecture 7, p. 99)**).

$$\hat{I}(X_i; X_j \mid X_S) = \sum_g \frac{N_g}{N} \cdot \hat{I}_g$$

average over the groups g

$$2N \cdot \hat{I}(X_i; X_j \mid X_S) = \sum_g 2N_g \cdot \hat{I}_g \;\approx\; \chi^2 \text{ with } G \text{ degrees of freedom}$$

G groups, one each

$$\text{average} \;\approx\; \frac{G}{2N}$$

a chi-squared variable averages its degrees of freedom

Our four friends, given the two others: $G = 4$, so about $4/1024 \approx 0.004$ for 512 days, against 0.0165 for the faintest true link 2–3.

More friends known, more groups, more fake information.

The loop's perfect notebook: all 16 counts

↳ Count = $512 \cdot \frac{1}{2}\cdot$(2 copies 1)$\cdot$(3 copies 2)$\cdot\left(\frac{1}{2}\cdot\right.$(4 writes 2's face)$+\frac{1}{2}\cdot$(4 writes 3's face)$\left.\right)$, each copy $\frac{7}{8}$, $\frac{3}{4}$ or $\frac{7}{8}$, else its complement.

day	count	day	count	day	count	day	count
HHHH	147	HTHH	4	THHH	21	TTHH	28
HHHT	21	HTHT	4	THHT	3	TTHT	28
HHTH	28	HTTH	3	THTH	4	TTTH	21
HHTT	28	HTTT	21	THTT	4	TTTT	147

Example: $n(\mathrm{HHTH}) = 512 \cdot \frac{1}{2} \cdot \underbrace{\frac{7}{8}}_{\text{2 copies 1}} \cdot \underbrace{\frac{1}{4}}_{\text{3 misses 2}} \cdot \Big(\underbrace{\frac{1}{2} \cdot \frac{7}{8}}_{\text{copies 2}} + \underbrace{\frac{1}{2} \cdot \frac{1}{8}}_{\text{misses 3}} \Big) = 28.$

Sixteen counts, each a product of named chances.

The collider's perfect notebook: 256 days

↳ Faces of friends 2, 3, 4. Count = $256 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \left(\frac{1}{2}\cdot(\text{4 writes 2's face}) + \frac{1}{2}\cdot(\text{4 writes 3's face})\right)$, each $\frac{7}{8}$, else $\frac{1}{8}$.

2, 3, 4	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
count	**56**	**8**	**32**	**32**	**32**	**32**	**8**	**56**

$$\Pr(3{=}\mathrm{H} \mid 4{=}\mathrm{H}) = \frac{56+32}{128} = \frac{11}{16} \approx 0.69$$

days with 4 = H

$$\Pr(3{=}\mathrm{H} \mid 4{=}\mathrm{H},\ 2{=}\mathrm{T}) = \frac{32}{32+8} = 0.8$$

friend 2 wrote T: 4 likely copied 3

Knowing friend 2 changes the guess about friend 3, once friend 4 is known.